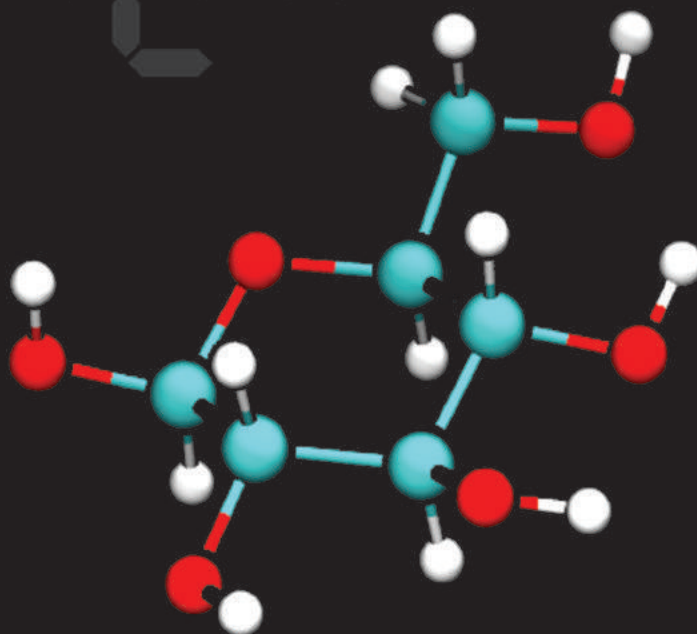
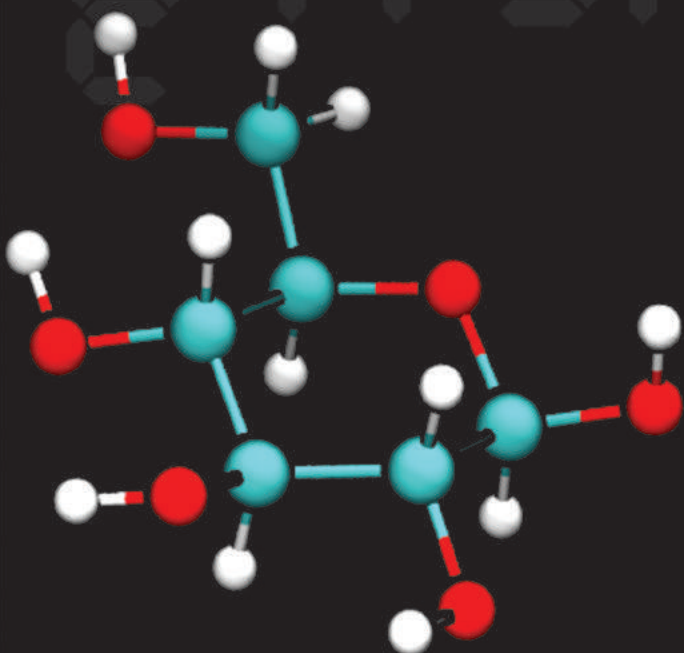


Symmetry



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C O N T A C T /kōn'tākt'/

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Dear Reader,

I am proud to present the Fall 2015 issue of the Berkeley Scientific Journal, based around the theme of “Symmetry.” In our discussion to decide this semester’s theme, one writer intelligently noted that much of science can broadly be characterized as a search for patterns and sense in a seemingly chaotic universe. However, a select few patterns can be characterized with that elusive term of “symmetry.” Symmetry is a concept that has fascinated humans for millennia. Ancient Greek philosophers like Aristotle famously held symmetry in the highest regard, believing symmetry to be one of the three “ingredients” to beauty. We see and prize symmetry in nature, from our appreciation of symmetric flowers to our natural attraction to symmetric images. And we artificially create symmetry, whether it be in the form of symmetric architecture or the symmetric form of famous literary works like Beowulf. This semester, our staff searched for elusive symmetry in the vast body of knowledge science has granted us over the years, and the results are sure to enthrall you and help you rethink symmetry in a new light.

In this latest issue, we explore symmetry through a combination of original review articles and interviews with Cal professors. We have an article relating symmetry to visual appeal [7], and the use of symmetry to treat chronic pain [10]. Also featured is an article discussing fractal geometry in tumors [18], and the use of computations to find a cure for cancer [12]. And our features section also includes a piece on the origins of chirality [1], and a discussion of ferroelectric thin films [3]. Our first interview features Hitoshi Murayama, a theoretical particle physicist, on the topic of supersymmetry, as it relates to the Standard Model of particle physics [21]. We also have an interview with chemistry professor Kenneth Raymond, whose lab researches supramolecular coordination chemistry, specifically the formation of symmetrical coordination clusters [35]. We are also proud to feature original academic research submitted by Cal undergraduates studying neuronal cell types [40].

Go Bears!

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Editor-in-Chief

S T A F F /stāf/

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ORIGIN OF CHIRALITY IN THE UNIVERSE

Shivaani Gandhi

Louis Pasteur was a French scientist of the 19th century with a passion to make a lasting contribution to science. In 1848, he postulated that a compound called racemic acid formed two distinct types of salt crystals due to some structural difference between the molecules the acid was composed of. He made solutions of the two different salt crystals and shone polarized light—light whose waves only move in one directional plane—through both, and found that each solution caused the light's direction to rotate an equal amount, but in opposite directions.

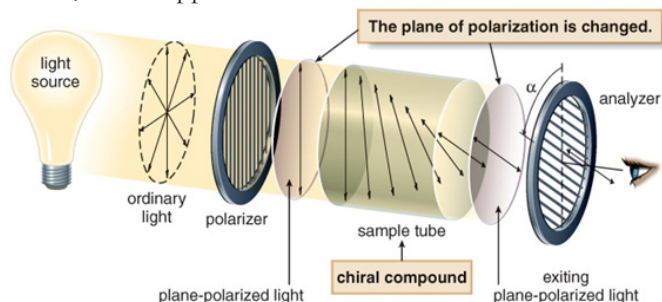


Figure 1.: Chiral compound rotating plane-polarized light.

His original postulation was later confirmed with the discovery of enantiomers, or molecules composed of the same atoms but with structures that are not the same, yet are mirror images of each other. (Newton, 2012) Such molecules are said to exhibit chirality. Sometimes, each enantiomer of a chiral pair is called “left-handed” or “right-handed,” since human hands are also non-identical mirror images of each other.

Chirality and optical activity—the ability of chiral molecules to rotate plane-polarized light—may seem like minor details that only chemists and maybe physicists would be concerned with. However, one simple difference in molecular geometry, even just switching the position of two atoms relative to the other ones in a molecule, can drastically change its properties. Some can be harmless, like in carvone: the right-handed enantiomer smells like caraway seeds, but the left-handed one smells like peppermint. Other times, the switch can have less benign effects: for the molecule thalidomide, the left-handed enantiomer relieves morning sickness, but the right-handed one induces birth defects (Schirber, 2009).

Taking these differences as examples, it may not seem surprising that chiral molecules in organisms exist almost exclusively as single enantiomers. In a lab, most syntheses will yield the left- and right-handed enantiomers in equal amounts, but in fact, most enantiomers in nature exist in the left-handed

form (Blackmond, 2010). Finding the origin of the increased left-handed and decreased right-handed concentrations is vital to understanding life. Finding out which chemical reactions sparked the creation of life on Earth can tell us what we might want to expect from life on other planets. Will they be left-handed molecules, like on Earth, or will their structures—and thus their characteristics—be completely different? Most scientists believe that homochirality is a precondition for life, with the argument that one hundred left- or right-handed gloves arranged in a sequence would have a well-defined structure, whereas a random mixture of both would be a mess if you tried to arrange them in a similar sequence (Schirber, 2009).

To attempt to account for the homochirality of biological molecules, scientists have created models that amplify an initial imbalance between the amount of two enantiomers in a system, which results in very large amounts of one enantiomer. These models are meant to model an amplification that, over time, may have resulted in the enantiomeric excess that we observe today. Proposed model systems include small initial imbalances in meteorites or other extraterrestrial sources, as well as random fluctuations in the physical and chemical environment that might account for a preference towards left- or right-handed molecules. Regardless of whether the excess was started off by chance or by design, an amplification mechanism is useful in understanding how the excess of left-handed enantiomers in our world got to this point (Blackmond, 2010). This article explores a few notable mechanisms that may explain the asymmetry in chirality in our world.

In 1966, David Cline, a professor at UCLA at the time, proposed a model by which organic materials, or carbon-containing compounds, were delivered to Earth from an interstellar source, such as meteorites (Cline, 1996). The homochirality observed in organic molecules may be a result of interactions with radioactive decay or supernova explosions in a dense molecular cloud. Chiral subatomic particles resulting from beta decay or effects of weak currents promotes asymmetric dominance, but this is on a very small scale. The amplification of this small asymmetry happens through a bifurcation process over a long period of time, or when a large number of these interactions happen over a short amount of time. Often, both may occur. The latter arises when radioactive decay or supernova explosions occur

very close to a dense molecular cloud, causing intense chiral impulses, in which a large amount of molecules in the cloud actually become chiral.

Another model developed this year at Harvard and presented at the 78th Annual Meeting of the Meteoritical Society in Berkeley this summer also cites space as the source of chiral molecules on Earth (Chan et al., 2015). Magnetite is an iron oxide mineral that is the most magnetic of all naturally-occurring minerals on Earth. Its magnetic properties allow it to catalyze the formation of amino acids found in certain meteorites. Electrostatic forces, or the attractions and repulsions between charged particles, occur between the surface of the magnetite, which contains the iron(III) ion, and the oxygen atoms in the amino acids. Sometimes, magnetite can take the form of plaquettes, or stacks of discs with a consistent change in their crystal orientation. They most commonly vary rotationally, and if they have certain key features, they can trigger an initial enantiomeric excess which can then be amplified by some mechanism to lead to the observed excess.

In general, as in both Cline and Chan's models, interactions between molecules and certain chiral objects, environments, or surfaces account for the formation of enantiomeric excess. In 2014, Morneau et al. proposed a model in which a prochiral molecule--a molecule that can be converted from achiral to chiral in a single step--interacts with a chiral surface and converts into a chiral molecule (Morneau, 2014). This model operates under the assumption that chirality exists somewhere in the world, and imparts chirality onto molecules that come into contact with it. Unlike the other two models mentioned, this model is mainly mathematical and calls on principles of quantum mechanics and mathematics for supporting arguments for this theory.

A new theory developed by Alexander Kusenko diverges from previous ones; according to Kusenko, an asymmetry that was present at the formation of the universe propagated the large amounts of asymmetry in chiral molecules that we observe today (Kusenko, 2015). For every 10 billion anti-particles, this theory proposes, there are 10 billion and one particles. Particles are slightly more energetically efficient than anti-particles, so anti-particles are converted into particles in order to be as energetically favorable and efficient as possible. This leads to an asymmetry in the ratio of anti-particles to particles, and the particle creation over time led to everything else in the universe. According to Kusenko, such asymmetry is necessary for creation, because if there were an even number of anti-particles and particles, upon collision, there would be a flash of light and little else. Thus, it may be that at the beginning of the creation of the universe, a tiny asymmetry which only took a few seconds to occur snowballed and led to atoms, which led to molecules, which led to the whole universe.

Overall, chemical models can very effectively attempt to account for the enantiomeric excess we observe today. However, there are plenty of theories that have supporting evidence and calculations, so there are no definitive conclusions on the origin of homochirality in our world.

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ENHANCEMENT OF FERROELECTRICS: STRAIN-ENGINEERED FERROELECTRIC THIN FILMS

Hongling Lu

In 1920, Joseph Valasek discovered that the polarization of the compound known as Rochelle Salt could be reversed by application of an external electric field. In effect, he was the first to recognize ferroelectric properties in a crystal. Ferroelectric materials are generally defined as those whose spontaneous polarization can be reversed through the application of an external electric field (Scott, 2007). (Fig. 1)

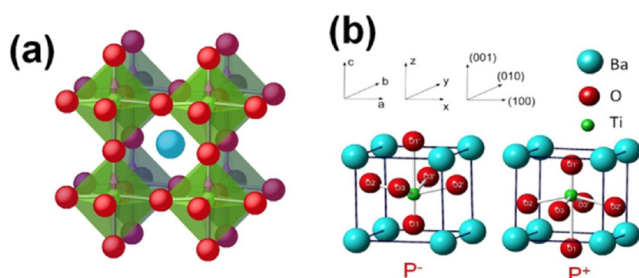


Figure 1: Perovskite oxides, of general formula ABO_3 with a pseudocubic structure, where A and B are two different cations, furnish many interesting ferroelectrics. The B-type cation is octahedrally coordinated with oxygen. In the example shown, $BaTiO_3$, it is the relative symmetry breaking displacement of the Ti atoms with respect to the O atoms which is responsible for the spontaneous polarization. $BaTiO_3$ has three ferroelectric phases: tetragonal, orthorhombic and rhombohedral. (Oxides. (n.d.).

Strain engineering is a strategy employed in exploring the special behavior of ferroelectric materials where a strained layer of ferroelectric epitaxially grown with respect to the crystalline substrate layer. (Eom et al., 2008) (Fig. 2)

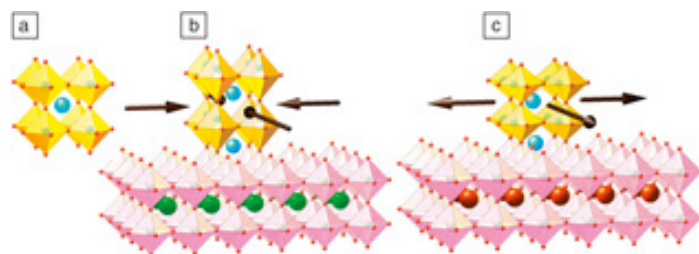


Figure 2: Before deposition, the structure is in an (a) unstrained state. When deposited on a substrate the energetic preference of the deposited atoms to follow the underlying substrate (epitaxy) can result in the film beginning in either (b) biaxial compression or (c) biaxial tension (Schlom, Chen, Fennie, Gopalan, Muller, Pan, Uecker, 2014)

This process creates a strained state. Under this strained state, the properties of the ferroelectric film differ greatly compared to the bulk ferroelectric, in a way that makes it suitable for use in applications such as memory storage on microchips. Before strain engineering, researchers used chemical alloying or doping to manipulate the properties of bulk ferroelectrics. A typical 0.1% strain--stretching of material to a length 0.1%

“Strain engineering is a strategy employed in exploring the special behavior of ferroelectric materials where a strained layer of ferroelectric epitaxially grown with respect to the crystalline substrate layer.”

At first, ferroelectric devices were restricted to bulk materials, but in the 1980s, thin-film, a technology utilizing extremely thin layers of material, was developed. The focus of ferroelectric study has since moved from bulk properties to its thin-film properties, especially with the promise of using ferroelectric materials in computer microchips, a type of thin-film technology.

Ferroelectric materials with spontaneous polarization due to the special arrangement of atoms in a lattice structure exhibit interesting pyroelectric and piezoelectric properties and have, therefore, attracted a lot of scientific attentions.

greater than the original length--will cause bulk ferroic oxides to crack. Epitaxial strain has the advantage of resulting in a more durable and ideally disorder-free film. Strain field is the electric field that results from a strained lattice structure, and in strain engineering, differences in lattice parameters between the epitaxial film and the underlying substrate are what create the strain of the overall thin-film material.

Past decades have seen a rapid progress in algorithms and calculations for analyzing thin-film ferroelectrics. Some of the methods include first principles calculations (determining electronic structure by solving Schrodinger's equation),

molecular dynamics (understanding how the molecules in the lattice structure move), phenomenological models based on Ginzburg-Landau theory (capturing salient features of the thermodynamic free energy for particular boundary conditions), and more, taking into account internal and/or external variables. (Schlom et al., 2014) These calculations are meant to predict electronic properties and structural characteristics of ferroelectric materials using computational methods. Predictions can then be verified and tested through experimental data to provide insight into ferroelectric investigations.

In strain engineered ferroics, the strain appearing in the film shifts the transition temperatures and can change the properties of the material such as the dielectric and piezoelectric constants and remanent polarization, or even can induce room temperature ferroelectricity in a non-ferroelectric material. (Haeni et al., 2004). Landau-Ginsburg-Devonshire (LGD) thermodynamic theory is the theoretical framework used for explaining and predicting equilibrium states and phase transitions in bulk materials. However, the order parameters (for ferroelectrics those are the electric dipole moment or polarization) used in regular LGD theory change considerably at the presence of the mechanical interactions between ferroic oxide and substrate. Thus, when applied to ferroelectric thin films with mixed mechanical conditions, the standard theoretical framework needs modification. A modified thermodynamic potential G is derived by an appropriate transformation. A minimization of such potential yields the equilibrium state of a film as a function of the temperature and misfit strain. One of the important predictions is proposed by Pertsev and Tagantsev using primitive free-energy theories. In the temperature–misfit strain phase diagram they generated, a new *ac* phase occurs with polarization lying between the *a* and *c* phase of BaTiO₃. However, the aforementioned methods are only applicable when the free-energy expansion is up to fourth order in order parameter (polarization). As pointed out in some later works on ferroelectric thin film, including works by Tagantsev in 2013, for perovskites ferroelectric a more common situation requires the potential to be expanded up to sixth order in polarization. Thus, higher order couplings should be taken into account. The missing higher order coefficients are calculated using *ab initio* methods. When high-order electromechanical coupling terms in thermodynamics of ferroelectric thin films are added and the new temperature–misfit strain phase diagram is drawn, the *ac* phase previously predicted disappears. This shows the power of *ab initio* modeling over primitive free-energy theories. (Scott 2007) (Fig. 3)

The thin-films discussed thus far have been homogeneous films with single composition. Novel behaviors can be predicted from modeling and observing experimental data for compositionally graded monodomain ferroelectric films. Compositionally graded ferroelectrics possess a spatial variation in chemical composition that breaks the symmetry of the system (Zhang et al., 2014). To account for the spatial

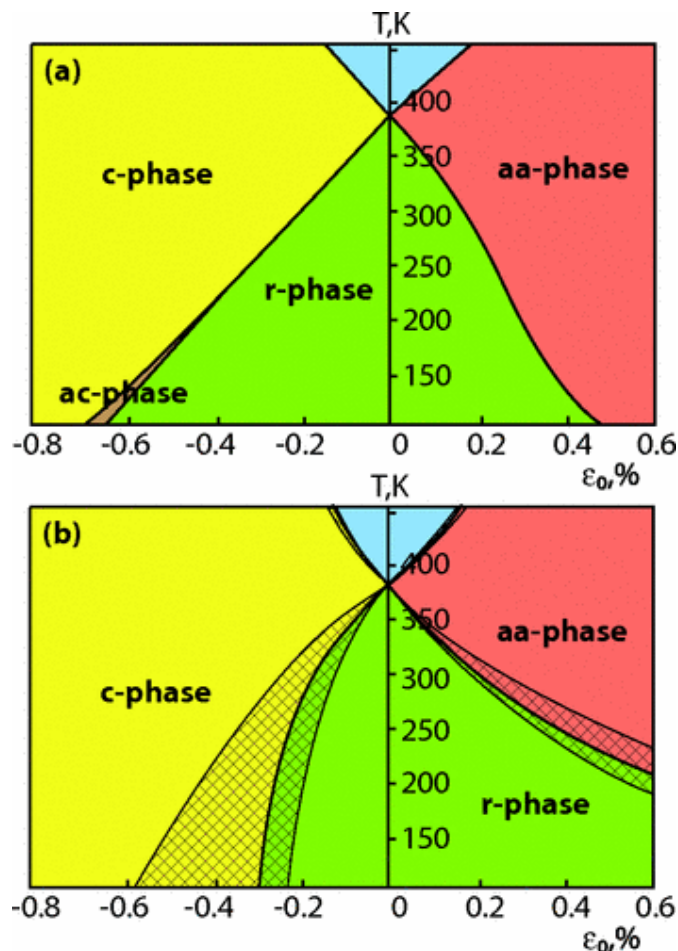


Figure 3: BaTiO₃ temperature-stress diagram with *ab initio* results (b) and results from Pertsev and Tagantsev in 1999 (a). (b) Developed with coefficient appended with the high-order coefficients. The hatched regions demonstrate the shift of the transition lines within the error bars of the coefficients. (Kvasov, & Tagantsev, 2013)

asymmetries, a new LGD-based thermodynamic framework is needed. Possible modifications include additional energy terms, including flexoelectric, gradient, and depolarization energies (Karthik et al., 2013; Ban, Alpay & Mantese, 2003; Catalan, Sinnamon, & Gregg, 2004; Eliseev et al., 2014). These gradient and energy tend to change the polarization behavior of the material by stabilizing a ferroelectric phase in the otherwise paraelectric composition. Some of the behaviors that are special to graded thin-film include: self-poling (Mantese, Schubring, Micheli, & Catalan, 1995), built-in potentials (Mantese, & Alpay, 2005), asymmetric or shifted hysteresis loop (Karthik, Mangalam, Agar & Martin, 2013), (Fig. 4) and the potential for geometric frustration (Choudhury, N., Walizer, L., Lisenkov, S., & Bellaiche, L. 2011). As a consequence, such systems are potentially important for a range of devices. (Mantese, & Alpay, 2005).

In 1976, Rotiburd predicted the polydomain phases of epitaxial layers theoretically. During the past years, experiments and simulations have proven the existence of polydomain structure on various crystalline substrates. Polydomain structure complicates the study by introducing

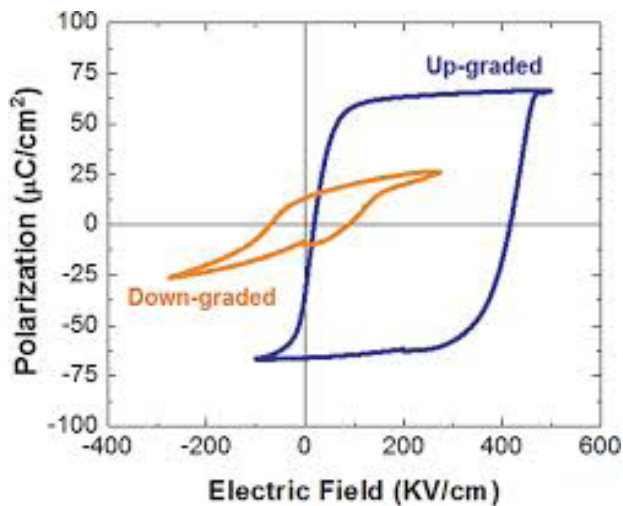


Figure 4: (Color online) Ferroelectric shifted hysteresis loops obtained at 1 kHz for compositionally up-graded and down-graded thin films. (Karthik, Mangalam, Agar & Martin, 2013)

internal mechanical stress caused by its inhomogeneity. The challenge can be solved by proposing new modeling to deal with existing stress at the film-substrate interface and the domain wall junctions. A rigorous nonlinear LGD thermodynamic theory has been developed for polydomain epitaxial films of perovskite ferroelectrics. The method makes it possible to determine polarizations, lattice strains, and mechanical stresses inside dissimilar domains forming dense laminar structures. Features of the theory demonstrate its great practical importance, because ferroelectric thin films have many possible applications in advanced microelectronic and micromechanical devices. (Koukhar, Pertsev, & Waser, 2001)

Thin-film technology made the development of high and novel electronic devices possible. The polarization of typical ferroelectric is reversed at a critical “coercive” field at about 50 kV/cm. In a bulk device, typically 1-mm in thickness, the voltage corresponding to this critical field is about 5-kV, which could not be put into mobile digital telephones. However, with submicrometer ferroelectric films, the voltage reduces drastically to less than 5V, permitting it to be integrated to microchips. (Scott 2007) Properties of ferroelectric thin-films are greatly enhanced by strained-engineering. With the distinctive electric field, thermal, and stress susceptibilities discovered by researchers through strain-engineering over the past decades, a range of devices such as transcapitors and transpondents have been developed. (Mantese, & Alpay, 2005)

Over the past decades, the study of strained-engineering constantly present exciting new perspectives in epitaxial thin-film science and encourage novel physical solid-state devices. This field of study is practically useful if we can better modulate the theoretical background and predict how to control and engineer the thin-film structure with desired physical properties.

“In strain engineered ferroics, the strain appearing in the film shifts the transition temperatures and can change the properties of the material such as the dielectric and piezoelectric constants and remanent polarization, or even can induce room temperature ferroelectricity in a non-ferroelectric material. ”

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Layout by Alexander Reynaldi

SYMMETRY AND VISUAL APPEAL

Shirley Shao

We are so used to the perfectly drawn faces in fairytale books or magazines, that when asymmetry appears in real life, we are startled. We notice people with lopsided smiles, or grins that are adorned with one dimple instead of two; we use eyeliner and layers of eyeshadow to cover up a size difference between two eyes. A face with a mole on one cheek but not on the other makes us pause, and we marvel over the perfect geometry of a beautiful movie star's symmetrical face.

Our eyes are highly equipped to detect bilateral facial symmetry, implying that facial symmetry must be important in some way. Symmetry in other objects can be detected by low-level visual mechanisms, but the detection of symmetry in one's face calls upon higher-level, more complicated visual mechanisms. Was there an evolutionary pressure to develop these higher-level visual mechanisms, so that we would better be able to detect facial symmetry? (Rhodes, 2005). Correlation does not necessarily indicate causation, but the signs of a strong positive correlation between symmetry and visual appeal seem apparent. In one study, photo manipulation was used to generate faces with varying levels of facial symmetry. The people who were asked to judge this series of faces consistently selected the faces with the greatest bilateral symmetry as most attractive (Rhodes, 1995). As given by Little's 2011 paper on facial attractiveness, symmetry can be defined as "the extent to which one-half of an object is the same as the other half (Little, 2011)." What, if anything, does symmetry have to do with facial attractiveness?



Figure 1: In Rhodes 1995 study, people were asked to rate how attractive these faces (each manipulated to display varying levels of symmetry) were. The most symmetrical

One logical proposal for the attractiveness of facial symmetry lies in the idea of "perceptual bias." By this principle,

people are predisposed to recognize certain stimuli in a certain way, preventing information from being processed in a wholly objective way (Gross, 2015). Human vision is built on bilateral symmetry – we have two eyes, one on each side of our face; the muscles associated with each eye's vision likewise display bilateral symmetry. Human vision can also be naturally divided into two fields – left, and right. When a visual point on one field can be matched to one on the other half of their field, the brain is able to process the visual information, and create a mental image with much more ease. Literally, a person with a symmetrical face is "easy on the eyes." Similarly, symmetry provides a template that allows a person's brain to construct at least half of an internal prototype that new information can be matched to. This rough outline that symmetry creates would also explain why people who have "average" looking faces are generally more attractive (Little, 2003).

By this vein of logic, faces that are symmetrical, but presented as upside down (with the mouth above the eyes, for example) should also be thought of as more attractive. However, this is not true – once the face we are viewing is inverted and therefore no longer upright, symmetry no longer increases the attractiveness of said face. An alternate explanation looks at the supposed genetic benefits conferred on a person with excellent facial symmetry. A mate can offer two different types of benefits – direct, and indirect. "Financial security" is an example of the former case, and does not necessarily measure a mate's genetic mettle. Rather, direct benefits confer a person and their offspring with an advantage in the present day; for example, an abundance of wealth or social status is immediately useful to a person and his or her offspring. Indirect benefits are subtler, and could entail long-term genetic benefits for future offspring. These then calls into question what, if any genetic benefits that facial symmetry could imply (Little, 2011).

Of note is that there are two different types of facial symmetry to consider. One is "fluctuating asymmetry" (hereon abbreviated as FA), and "directional asymmetry" (known as DA). Directional asymmetry is asymmetry that takes into account the prevalence of hemi-face dominance; in these situations, the line of symmetry splitting the face of a person who exhibits directional asymmetry will not be in the center middle of his or face. Rather, human faces are often larger on the right side, and this asymmetry is exploited when one is trying to convey different states of mind. For example, people are wont to show more of the right side of their face when they want to hide their emotions (Simmons, 2004).

Fluctuating asymmetry, or its absence in a person's face, is a better indicator of a person's health. This is what we usually think of, when we consider "asymmetry" in a person's face affecting how attractive they are. Fluctuating asymmetry describes variance over the line of symmetry splitting a person's face. They encompass variations on top of directional asymmetry and result from a person's experiences during early development. If a person's immune system is weak, and unable to sufficiently defend against outside stress, a person's face will begin to exhibit greater deviations from perfect symmetry. Therefore, larger amounts of fluctuating asymmetry can be a reflection of instability during development (Simmons, 2004).

Incidentally, males tend to exhibit higher amounts of fluctuating asymmetry, because testosterone represses the immune system during development. This hormone weakens the body so that it is more susceptible to

parasitic infections that would prevent perfect facial symmetry from forming. Greater amounts of testosterone are also related to increased rates of prostate cancer. Yet, testosterone also makes the development of secondary sexual traits possible, traits that are very often thought of as attractive in men (Rhodes, 2003). There then appears to be a trade-off between the good health of symmetry, and those secondary sexual traits – unless a person's immune system can superbly defend against parasites and other environmental stresses, even when weakened by testosterone. In the animal world, male peacocks show off their striking plumage as a way of indicating that they can survive and thrive in spite of an attribute that should have evolutionary detriments. For humans, secondary sexual traits can be like the peacock's striking tail – an indication of health so robust and well-adapted, that their body can compensate for the costs of suppressing the immune system. This is the "immunocompetence-handicap hypothesis," wherein a person who can bear a higher parasite burden, can also display greater expression of secondary sexual traits (Rhodes, 2003).

However, we must not neglect the genetic factors that can increase male facial masculinity (masculinity that results from greater expression of secondary sexual traits). There is a widespread belief that a facially masculine man should be able to offer greater benefits (genetic or otherwise) to their offspring, but such a postulate has not been rigorously researched. Whether or not increased masculinity actually offers an evolutionary advantage should be considered with regards to the population as a whole. Since "masculinity" (again, defined here as secondary sexual traits associated with males) has a genetic factor, males with more masculine faces will also have sisters who are facially more masculine; there is nothing contradictory about this statement, given that the amount

of testosterone a person produces is related to their genetic make-up. Facial masculinity in a male may or may not make the male more attractive to their prospective mate, but their sisters are, on the whole, viewed as less attractive (Lee, 2014). The "immunocompetence-handicap hypothesis" would have one choose a man with more masculine features for the perceived indirect benefits for one's offspring, but the same "masculine" traits would give one's daughters a reproductive disadvantage. Interestingly, men with feminized faces are sometimes found to be more attractive for said "feminine" features, implying that females who are regarded as more attractive for their "feminine faces" would not have brothers who have a reproductive disadvantage (Little, 2011).

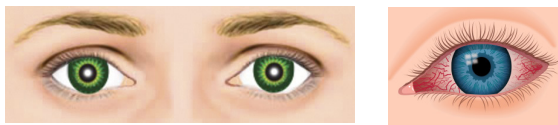
Meanwhile, in a 2004 study by Koehler, researchers found that high facial femininity was associated with higher body symmetry. In turn, increased levels of

high facial femininity and high body symmetry were associated with overall attractiveness, implying that body asymmetry was also an indicator of developmental instability (Koehler, 2004). Likewise, feminine faces may be more attractive, but they are not healthier than their supposedly less "feminine" peers. This further suggests that the evolutionary benefits that could be conferred by seeking attractive facial traits is tenuous (Rhodes, 2003). It may be more effective, then, to separate symmetry away from concepts of masculinity and femininity as a category of attractiveness (Little, 2011).

Perhaps more important to consider is that, while the reasons for and existence of a biological preference for facial symmetry are debated and uncertain, the strong correlation between healthy faces and greater attraction is clear. Here, we think of a study done on sclerae, the typically white covering around the eyeball. As previously mentioned, humans have two eyes; when we cry, normally white sclera become pink or red. White scleras are a reflection of normality, are healthier. When one eye is red, the face exhibits asymmetry; when both eyes are white, or both eyes are red, the face exhibits symmetry. People in the study had a negative reaction to seeing two red eyes, a better reaction to seeing one red eye and one healthy white eye, and the best, most positive reaction to seeing two white scleras. In relation to the symmetrical red eyes, the asymmetry of having one white eye was positive; however, in the end, the normality of two healthy white scleras was preferred above all. In this situation, the color of the sclera was not an indication of genetic makeup, but a nonpermanent reflection of a person's current mood. Symmetry in this case would not necessarily confer an evolutionary advantage of any sort, and instead, increased health is the main factor that determines overall attractiveness (Provine, 2013).

"Symmetry may not necessarily confer an evolutionary advantage of any sort, and instead, increased health is the main factor that determines overall attractiveness."

Indeed, in spite of a lack of perfect symmetry, faces can, and are still found attractive. A lopsided smile, indeed, may only add to the charm of a person; Marilyn Monroe's beauty mark lent her face asymmetry but invited imitation, not disgust. In the end, it would appear that the general human predilection towards good health holds true, even if the cost entails an amount of asymmetry.



Figures 2 and 3: White vs red sclera

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Layout by Rhea Misra

IT'S ALL JUST SMOKE & MIRRORS

Liza Raffi

Ever since the invention of the mirror, man has used his reflection as a tool for composing and constructing body image. Research into potential therapies for a novel pain syndrome has taken this age-old practice a step further by exploring whether a mirror's virtual image can trick the brain into integrating into that body image a limb that no longer exists, allowing one to control and exercise it. Phantom limb pain, which affects over 85% of upper and lower extremity amputees, is at once clinically well documented and etiologically confounding. Written off in the past as a psychological maladjustment to the emotional and practical upheaval resulting from losing a limb, phantom limb pain has been increasingly examined in the last two decades from a neuroscience perspective geared toward understanding its potential physiological underpinnings. Mirror Visual Feedback (MVF) therapy, developed by Dr. Vilayanur Ramachandran, has made preliminary strides both as a therapeutic option for

treating phantom limb pain and as a research tool for teasing apart the various causal mechanisms that could be at play. In this article we will take a look at the methods and merits of MVF, as well as interesting insights into the plasticity of adult neural networks that have been uncovered by Ramachandran and others' work.

While the sensations experienced as pain in a phantom limb vary and may depend on numerous factors – the limb removed, the length of time spent with injury before its removal, use of a prosthesis – sensations of cramping, stretching, paralysis, and intermittent sharp pain are among those most frequently described. Medications, such as narcotics and antidepressants, have had some success in slowing or preventing the uncontrolled pain signals but come with a wide array of side effects, including risks of addiction and dependence. Nerve blocks, slightly more invasive, use the injection of a local anaesthetic or corticosteroid between



“One patient was even able to eliminate awareness of a phantom limb altogether -- what Ramachandran calls the first amputation of a phantom limb!”

the spine and the amputation site to sever the pathway of the pain signal. If the phantom pain is debilitatingly severe one may even opt for surgical revision of the stump site. Unfortunately, effectiveness of any of these methods is largely unproven, with response rates rarely exceeding that of placebo treatments (Brodie et al). Into this scene entered neurologist Vilayanur Ramachandran and his mirror box in 1996, the beginning of MVF therapy.

Ramachandran and his team studied 10 upper limb amputees. The so-called “virtual reality box” is a simple and inexpensive construction. A mirror is placed vertically on the table so that the mirror reflection of the patient’s intact hand is ‘superimposed’ on the perceived position of the phantom. With the phantom arm visually resurrected, the patients are then guided through various exercises. In six patients, movement of the intact hand while visualizing the virtual limb resulted in the same kinesthetic sensation (the sensation of one’s body movement) in the phantom hand. Ramachandran notes that this occurrence was often to the surprise of the patients, several of whom attested that having volitional control over the phantom was pleasant in itself. Further, of the five patients who had experienced painful clenching spasms prior to the study, four had the spasms relieved when the mirror was used to aid in “opening” the phantom hand. One patient was even able to eliminate awareness of a phantom limb altogether -- what Ramachandran calls the first amputation of a phantom limb!

“If motor commands and the resulting visual/proprioceptive feedback are contradictory (as they are in amputees) the arm becomes immobile; however, restoring the feedback (e.g. using the virtual reality box) revives mobility in the phantom.”

What could be aligning to change the nature of the phantom? Ramachandran postulates that these mirror images provide key feedback to the parietal lobe, the region of the brain which integrates various streams of information to construct a dynamic body image. The parietal lobe processes signals from the motor/pre-motor cortex that indicate intention to move a body part, and in normal cases it also monitors performance of those actions using visual and proprioceptive feedback. Proprioception is one’s sense of where their limbs are in space based on a network of receptors in the skin, and is a component of the kinesthetic sense described above. If motor commands and the resulting visual/proprioceptive feedback are contradictory (as they are in amputees) the arm

becomes immobile; however, restoring the feedback (e.g. using the virtual reality box) revives mobility in the phantom. More than anything, Ramachandran was amazed by what appeared to be the systematic and topographical referral of sensation from patients’ normal limbs to their phantom limbs. It appeared that within three weeks, there had been the rapid emergence of precise and organized pathways linking the two cerebral hemispheres in the adult brain. Due to the

“When the right hand is amputated [ipsilateral] input may become either disinhibited or progressively strengthened so that touching the left hand evokes sensations in the [phantom] right hand as well.”

implausibility of axonal growth across those distances so quickly, Ramachandran hypothesizes that the exercises enhance pre-existing commissural connections. While we know that sensory input onto, say, one’s left thumb is projected onto the right hemisphere, Ramachandran further suggests that it is also relayed to symmetric locations on the left hemisphere. This latent input may ordinarily be too weak to express itself, but when the right hand is amputated this input may become either disinhibited or progressively strengthened so that touching the left hand evokes sensations in the right hand as well. This finding, strengthened by fMRI and PET images published in a later paper published by Ramachandran, is a surprising indication of neural plasticity in adults and possible remedial avenue for stroke victims suffering from paralysis as well as plain-plagued amputees.

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Layout by Rhea Misra

COMPUTING THE CURE TO CANCER

Kirk Mallett

The approaching future of health care is uniting humans and machines to tirelessly attack the most challenging diseases. Computers are essential to conquering recalcitrant diseases through extreme precision and awareness. Diseases such as cancer, that are predominantly controlled immunologically, genetically, and epigenetically necessitate individualized prognosis and treatment. During the next

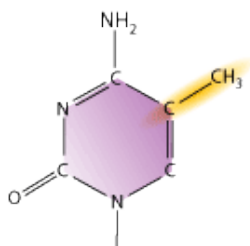
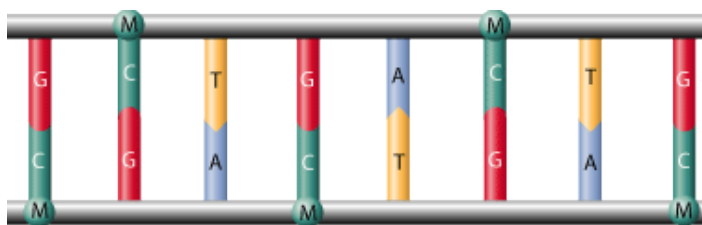
Personalized cancer treatment requires large amounts of patient specific data. The central dogma of cancer progression is the buildup of mutations in the genetic sequence of certain genes. These oncogenes are especially critical in bestowing our cells with the tools and behaviors of cancer. Genetic analysis informs your doctor about what type of cancer you face, indicating the treatments more likely

“In epigenetics, there is an important asymmetry, that nearly your entire body is made of cells that are genetically identical, yet irreversibly differentiate into specialized roles.”

several years, we will see the coevolution of medical science and machine intelligence to provide such personalized care. In the following, we will see that to meet the instrumental and computational challenges of cancer and other diseases will require substantial progress beyond what presently occurs in oncology labs and clinics. By assessing cancer as if it were essentially physiological information, we see the importance of genetics and epigenetics during diagnosis and treatment. Then we look at the role of biomarkers in improving the state of practice in the lab and clinic, all the while emphasizing the imperative to work alongside machine intelligence.

to defeat your tumor, which grows wildly in your organ. In some approaching year, the specific cells betraying your body will succumb to a treatment tailored uniquely to those cells. Few other cells will be harmed, dramatically minimizing side effects. Yet this specificity cannot be solely based on genetic information, which are the instructions on the construction of protein. Proteins are the nanomachines operating the complexities of both your healthy cells and your cancerous cells, and we know these cells are different by looking at their genetics. Genetics best informs us about what variants of proteins your cancer might express, and about how they differ from healthy cells. Genetics does not indicate at what levels proteins are expressed, if at all.

“The larger a dataset is, and the more sophisticated the analysis becomes, the much greater the time required to process that data.”



DNA methylation is the addition of a methyl group (M) to the DNA base cytosine (C).

“Graphs, in the form of

Hidden Markov Models

(HMMs) underlie the symmetry

involved in a lot of biomedical

analysis.”

Transcriptomes, methylomes, and other epigenetic information, not genetics, tells us which proteins are being expressed at what levels and how they are being regulated (Dancey). Transcriptomes of RNA are difficult to access from cells, but will one day compliment genetics in deciphering what to target in your unique cancer. The pattern of DNA methylation and histone modification in tumorigenic cells could map out potential regulatory targets. In the future, there may be some drugs that stop the expression of critical proteins in your cancer. Epigenetics regards the regulation of genetic expression, when and which proteins are produced in a cell, and the variation of such regulation between different cell types and cells of the same type in different environments. In epigenetics, there is an important asymmetry, that nearly your entire body is made of cells that are genetically identical, yet irreversibly differentiate into specialized roles. Scanning through cells of every tissue we see a symmetry about their potential to produce any protein and perform any cellular role. This symmetry is broken across the axis of expression, the potential to be anything is suppressed in order to create specialized cells. Within tissues and between cells there is differential regulation of protein generation; we are a unified body of cloned cells that distinguish themselves solely through their varied expressions.

“...when available, a doctor

today can sometimes, and to

a limited extent, formulate a

personalized treatment for a

cancer patient.”

Cancer also differentiates itself, but takes differentiation a step beyond expression; cancer is a polyclonal network of highly interdependent cells (Parsons). To conquer your body's particular brand of cancer, not only must changes in genetic sequences and expression levels be monitored and understood, this data must be isolated from different clonal populations in a highly heterogeneous tumor. To acquire this data requires a large assortment of tools and a robust biomedical industry. Sophisticated mathematics and continued growth in computational capacity will process this data until it presents insight and actionable information. If you are fortunate enough to outlive vascular diseases, you will someday be asking your physician about a lump or pain on your body somewhere. With the onset of cancer, you will be mollified by the quality of information your doctor has on your physical state, and by the variety of treatments that can be tailored to your specific tumor. You may or may not see

“So data parallelism and ILP

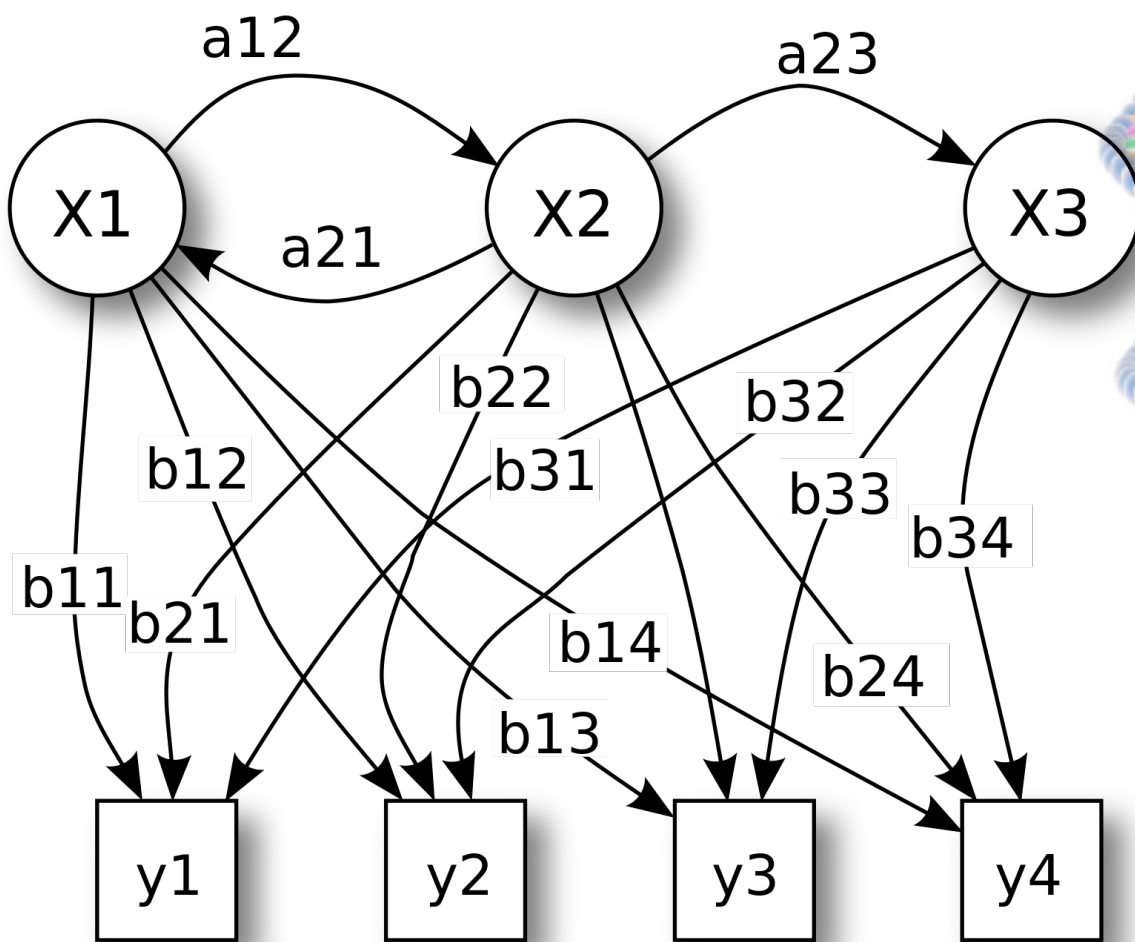
both derive from the fact that

complex genetic patterns can be

based on simple premises.”

yourself as a unique individual, but your doctor will know your cancer uniquely. However, this will require datasets of great size being analyzed at tremendous speeds. The larger a dataset is, and the more sophisticated the analysis becomes, the much greater the time required to process that data. Personalized treatment is therefore impossible without exploiting inherent symmetry through computationally efficient mathematics.

Graphs, in the form of Hidden Markov Models (HMMs) underlie the symmetry involved in a lot of biomedical analysis. Interpreting sequences of DNA (Meng), DNA methylation patterns (Lee), or regulatory motifs in DNA (Wu) will become dominated by HMM methods. HMMs are graphs of observables, say a position in a DNA sequence that can be an A, C, G, or T nucleotide. A position in a particular genetic motif may have a 10% chance to be an A, a 33% chance to be a C, 42% chance to be a G, and a 15% chance to be a T. The motif itself may have a 30% chance of occurring after another motif 1a, a 24% chance of occurring after a motif 1b, and a 20% known chance of occurring before a motif 3. So we see that a particular position in a particular motif has a probability of being an A, T, C, or G, and this probability is dependent on where the nucleotide might be in which possible motif (Meng). The most probable description



Typical Hidden Markov Model (HMM)

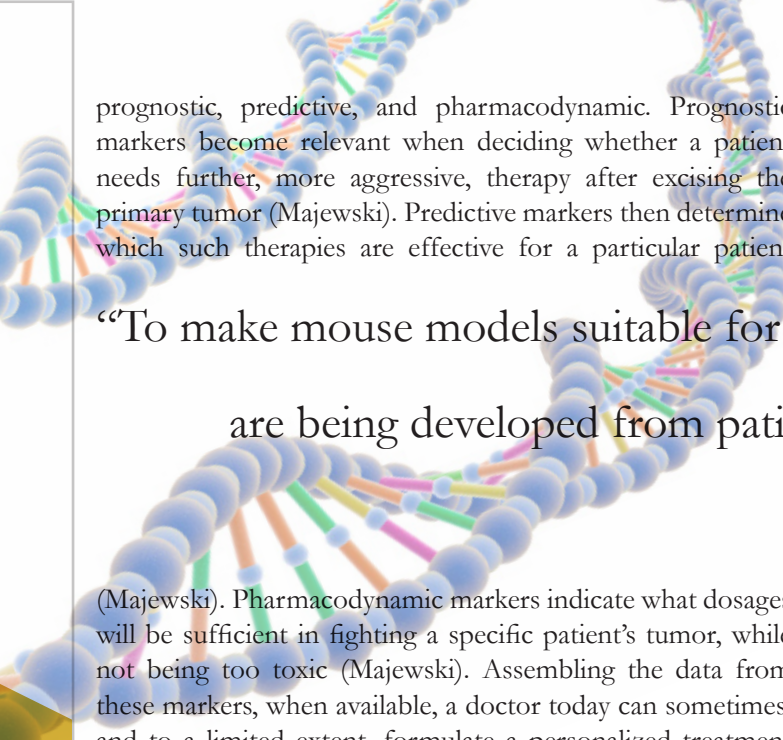
of the DNA sequence then becomes whichever arrangement of motifs is most likely to produce the observed sequence. This description is of a lower fidelity than in modern models, which should also include analysis between observables larger than a single nucleotide (Lee).

To train a model to recognize motifs correctly, the model must try many possibilities. This is computationally taxing, but there are inherent symmetries that speed the computations up. Though there are only four kinds of nucleotides to consider, they form very long combinations that can be as complex as they are long. These sequences can be nonrepeating, and potentially highly interdependent, in theory. HMMs avoid this problem by modeling each position in a sequence so that it has only one dependency, the nucleotide in the previous position (Lee). Since every one of the billions of nucleotide positions are symmetric in their limited dependencies, their computation can be distributed across many processing units (Meng). For the same reason, the storage and movement of nucleotides, and the instructions for analyzing them, can be efficiently managed (Meng). A second source of computational symmetry is called Instruction Level Parallelism (ILP), which HMMs elicit through their basic operations (Meng). While the properties of an HMM can be very difficult to prove, the model uses fundamentally

“...despite very large datasets on genetics and expression, very little insight has emerged from analysis of that data, limiting the progress of oncology.”

simple operations, such as multiplying a small list of numbers together. Though these operations must occur many billions of times, one operation can be simultaneously performed on several positions in a sequence or across several steps in a chain for a single nucleotide. So data parallelism and ILP both derive from the fact that complex genetic patterns can be based on simple premises.

There are three stages of cancer treatment that are improved with information on genetic (or other) markers:



prognostic, predictive, and pharmacodynamic. Prognostic markers become relevant when deciding whether a patient needs further, more aggressive, therapy after excising the primary tumor (Majewski). Predictive markers then determine which such therapies are effective for a particular patient

closely related cells begin to dominate the tumor, and some of those will be or become resistant to the treatment. What we need are improved ways to identify and target families of tumorigenic cells.

“To make mouse models suitable for personalized medicine, mouse avatars are being developed from patient derived tumor xenografts.”

(Majewski). Pharmacodynamic markers indicate what dosages will be sufficient in fighting a specific patient’s tumor, while not being too toxic (Majewski). Assembling the data from these markers, when available, a doctor today can sometimes, and to a limited extent, formulate a personalized treatment for a cancer patient. Personalized cancer therapy has had some reserved success, such as targeting HER2 in breast cancer, BCR–ABL translocations in chronic myelogenous leukemia (the less common, less aggressive leukemia), the EGF receptor in lung adenocarcinoma, and BRAF mutations in melanoma (Tyson). These targets tend to be fusion proteins arising from mistakes in the separation of chromosomes (Rodriguez-Antona). In horrifying irony, by targeting tumor cells that carry these mutations, selection for resistant cells occurs among close genetic variants of targeted cells (Tyson). These resistant cells perpetuate the tumor despite treatment. In other words, by killing only the most tumorigenic cells,

Only half of recently approved drugs have known biomarkers associated with them to indicate whether a patient might respond to the drug, what extent that response might be, or when dosage becomes toxic (Rodriguez-Antona). This means that despite very large datasets on genetics and expression, very little insight has emerged from analysis of that data, limiting the progress of oncology. Inadequate biochemical tools and techniques in the laboratory is a part of this retardation. Insufficient biomarkers, their assays, and the means for their bioconjugation limits information on molecular pathways (signal cascades) in cancers. Costs and risks have continually increased, and drug development slowed, for the discovery and refinement of new drug targets (de Castro). The quest to make personalized cancer treatments robust and routine has consequently faltered. Moreover, without critical information on drug pathways, side effects cannot be predicted. Long term concerns of genetic toxicology, how acute chemo- and radio- treatment affects genetic stability, remains unknowable in healthy tissue. Nevertheless, hope is high that progress will triumph over these challenges. In addressing these shortcomings and revolutionizing treatments, very large datasets are expected to be assembled and must be processed.

“The growth of this data implies the need for Natural Language Processing (NLP) agents that inform doctors about potential surgical outcomes and responses to drug and radiological treatments.”

One source of this data will come from mice, which, besides being genetically similar to humans, are relatively easy to genetically alter. Mice also have a short gestation period, and are cheap to house. However, mouse models have lacked the clonal and signaling heterogeneity of human tumors; they are simply too simple to model our diseases. To make mouse models suitable for personalized medicine, mouse avatars are being developed from patient derived tumor xenografts. Immunodeficient mice are transplanted with a biopsy sample from a patient’s tumor. From this, a mouse line is raised and used to test various cancer therapies, looking for ideal agents and dosages for an individual patient (Malaney). This data can be combined with whole-exome sequencing that can also be performed on biopsy samples (Rodriguez-Antona). This combined approach has recently been trialed, successfully treating thirteen patients. Six patients saw partial remission and the other seven experienced disease stabilization, having no progression of the tumor (Garralda). In eleven cases the

avatar model mimicked the patient response (Garraalda). Over ten million codons of selected genetic regions, from each patient's tumorous and healthy samples, were analyzed to find only an average of 45 mutations (Garraalda). Apparently, observing all the significant mutations of a cancer requires a high degree of fidelity. Yet even this level of scrutiny is not considered sufficient for general application of personalized cancer treatment (Parsons). We see the need for detailed yet efficient analysis of the millions of biopsies per year.

The most common and most important source of information in the clinic and in the medical science laboratory, today, is text based documentation and communication (Jensen). The growth of this data implies the need for Natural Language Processing (NLP) agents that inform doctors about potential surgical outcomes and responses to drug and radiological treatments (Jensen). NLP agents interpret and make sense of normal human language, such as English or Chinese, usually as blocks of text rather than spoken phrases. IBM is developing such an expert agent, a medical assistant based on their Watson project. IBM's product relies on numerous techniques to analyze text for patterns meaningful to doctors. In Watson, text is turned into nodes and lines on graphs. Phrases that emphasize nouns are designated as branch points, and phrases emphasizing verbs become the connections between those branch points (Kalyanpur). These phrases are created by Watson from simpler linguistic features, like parts of speech, and from dictionaries (Kalyanpur). An extension to Watson, called WatsonPaths, breaks a question into a set of smaller questions after Watson provides its top rated responses to the original question (Lally). WatsonPaths also asks the doctor questions, and utilizes the doctor's response to improve Watson's answer (Lally).

Microenvironment, developmental state, cell type, and other factors modify the expression and activity levels of hundreds of relevant molecular components in each tumor (Chin). The varieties of cancer with their diversity of genetic mutations compound the number of assays and molecular tools needed in both the laboratory and clinic (Chin). Great concentration of resources will compel progress in these areas, and to interpret and create value from the amassing data demands proportionate computational advances. There is a glimpse of the future today in the development of IBM's Watson and its successor. Beyond the symmetries in genetics and epigenetics, there is more symmetry to exploit in immunology, and endocrinology (Kolch; Brock; Melero). These symmetries become apparent in the theory and computation of modeling the complex interactions ubiquitous in those domains. By exploiting symmetries of physical and combinatorial structures that are medically relevant, several critical problems have become tractable. Simplifying the identification of disease relevant genes in an individual tumor is one instance. Symmetry is integral to simplifying and solving other principle challenges, like mapping the nearly inscrutably

attenuated regulatory pathways of disease progression. These eventual triumphs await long tribulations of discovery, invention, and investment that will require greater integration of health industries and consumers (Fagnan).

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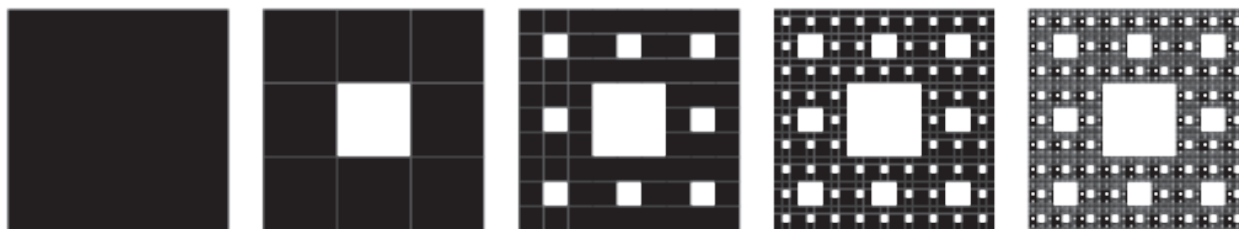
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Layout by Kara Turner

SYMMETRIC PROLIFERATION:

AN EXAMINATION OF THE FRACTAL GEOMETRY OF TUMORS

Rachel Lem



Sierpinski carpet model of plane fractals

From flowers to faces, nature is abound with symmetry. Most natural objects tend to form according to patterns, a tendency which mathematicians readily exploit in order to create theoretical models of the world around us. The height of a cliff, for instance, is modeled by a one-dimensional line; a snake's path through the sand is modeled across a two-dimensional surface; a block of ice is modeled as a mass extending into three dimensions. But what about objects in between dimensions? In fact, between the 1-D and the 2-D there exist objects known as fractals. These mathematical objects are infinitely self-similar, which means that upon magnification of a certain part of a fractal, one sees the figure of the overall fractal, and so on unto infinity. Self-symmetry

"Picture, for example, water trickling through only the most loosely packed areas of soil in a pot; in the same way, the blood vessels of a tumor grow into the weakest areas of the tissue around it."

allows a fractal to have fractional dimensions because it is not purely linear--the border of a fractal cannot be traced--but this lack of boundedness also means that the fractal never encircles a defined area. Imagine a tree whose branches branch out infinitely, or a snowflake with six tips, each of which looks like the original snowflake, and so on and so forth.

Clearly, in fractal models, as in all models of nature, there is a difference between the mathematical and the natural. Natural fractal objects are composed of discrete units and are not infinitely divisible--the fractal pattern must end somewhere, or else, for instance, one might find tiny branches at the cellular level of a tree branch. To correct for this quality,

scientists define natural fractal objects as having statistical self-similarity, or when "the statistical properties of the pieces [of an object] are proportional to the statistical properties of the whole" (Grizzi et al., 2008).

In the human body, statistically self-similar models are most commonly applied to branching structures in the lung and in networks of blood vessels. The latter application has had particular importance in medical studies of cancer, as there is evidence that understanding the fractal geometry of tumor vasculature may aid in identification and targeted treatment of cancers.

Tumor vasculature can in fact be described by a fractal model, and is often distinguished from normal vasculature by either an abnormally high or abnormally low fractal dimension (Zook and Iftekharruddin, 2005). An object's fractal dimension is a constant between the integers 1 and 2, and might be described as how 'proliferative' the object looks; i.e., an object with a higher fractal dimension looks more like an object with true area than like a curve. Baish and Jain observed that blood vessels in the tumors of mice had higher fractal dimensions than the mice's normal arteries and veins, claiming that "the fractal dimension quantified the degree of randomness to the vascular distribution, a characteristic not easily captured by the vascular density" (Baish and Jain, 2000). Moreover, the researchers noted that tumor vessels tended to be more twisted than normal vessels, having "many smaller bends upon each larger bend" (Baish and Jain, 2000). They also found that the way tumor vessels grew and branched closely matched a type of statistical growth called invasive percolation. In invasive percolation, a substance moves through a medium which has varying degrees of strength, penetrating the weakest areas of the medium and thus branching out to form a network. Picture, for example, water trickling through only the most loosely packed areas of a pot of soil; in the same way, the blood vessels of a tumor grow into the weakest areas of the tissue around it. On the other hand, normal capillaries are traditionally modeled by the Krogh cylinder model, which assumes that the capillaries are straight, relatively spaced, and only reach a cylindrical volume of tissue immediately

surrounding each linear capillary. Given that the Krogh model idealizes even the most organized vasculature, it is clear that a statistical fractal model is better suited for tumor vasculature, which lacks arterioles, venules and capillaries, and whose irregularly shaped vessels often do not even interconnect (Folkman, 2002).

In the early 1990s, a series of studies concluded that

“Knowing where the tumor vasculature reaches is akin to knowing where in the tumor the treatment can reach.”



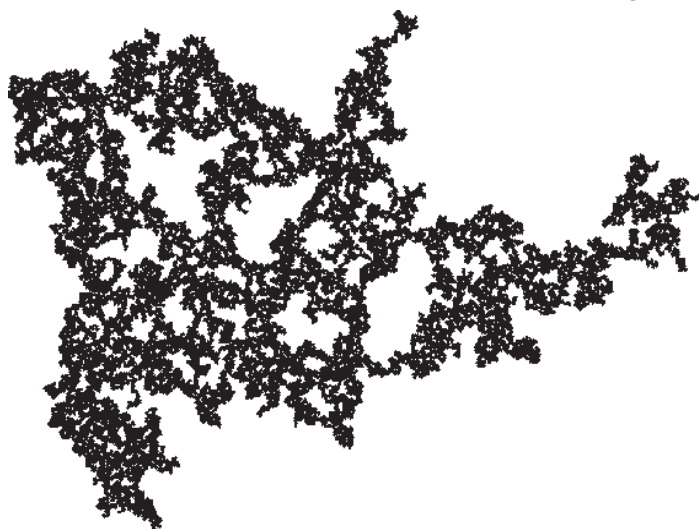
Vascular tumor

tumor microvessel density (MVD), or the degree to which the cancerous tumor has established its own vasculature, is associated with metastasis of that cancer (Folkman, 2002). Since then, knowledge of tumor vasculature has been applied

in attempting antiangiogenesis, or prevention of blood vessel growth, as a proposed way to control tumor growth. However, antiangiogenesis has historically had limited success. Some scientists hypothesize that the irregular geometry of tumor vasculature--given by an abnormal degree of self-symmetry--results in two problems that mirror each other. First, while disorganized vasculature makes it difficult for tumor cells to receive nutrients, it also makes it harder for drugs targeting the tumor to reach a good proportion of the tumor. (Chauhan et al., 2012) Second, if an antiangiogenic drug does succeed in spreading to most of the tumor, the tumor might instead develop relatively normal vasculature that then allows for nutrients to be better transported to tumor cells, speeding up the growth of the tumor.

Nevertheless, understanding tumor vasculature is still useful since many cancer treatments are affected by drugs which flow through the tumor's blood vessels. Knowing where the tumor vasculature reaches is akin to knowing where in the tumor the treatment can reach. To test this reach, Baish and Jain performed another study in 2012 in which a tracer transport model was coupled to a model of blood flow based on invasive percolation--essentially, the researchers created a fractal model of tracer movement through a tumor, by which the tracer represented a potential drug. This model predicted “highly heterogeneous transport in the tumor,” which the researchers deemed “clinically significant because some ‘out of the way’ regions of tumor may receive low concentrations of [the drug]” (Baish and Jain, 2012). In an article examining medulloblastoma in children, Grizzi, Weber and Di Ieva also support the use of a fractal model for tumor vasculature,

and go even further to say that this fractality implies a level of irregularity in blood vessel organization that renders MVD a poor measure of the degree to which the tumor has established its vasculature. The article concludes that greater



Dynamic of fluids in porous media and critical percolation phenomenon

focus should be placed on modeling tumor vessel networks as fractal objects, so that scientists might better understand where in the tumor the drug cannot reach, and possibly devise drug delivery methods that work around this difficulty.

Interestingly, Brú et al. used a fractal model to discount antiangiogenesis entirely as a treatment for cancer. These researchers focused their attention on the growth of the tumor as a whole, observing fractal geometry in the way the cells proliferate around the edge, or the contour, of the tumor. According to their article in Biophysical journal, such fractal growth is an indication that the tumor always maintains a layer of actively proliferating tumor cells about its contour. Their article challenges the belief that decreasing vascularization--i.e. antiangiogenesis--to effect cell necrosis could effectively combat tumor growth, on the basis of the idea that it is not poor vascularization that prohibits cell proliferation toward the center of the tumor, but rather “pressure effects” (Brú et al., 2003). Thus, poorly vascularized tumor cells could hypothetically still proliferate actively, as long as they are near the contour of the tumor, where pressure is lower.

In sum, examination of the fractal geometry of tumors reveals a similarity between the way fractals are infinitely proliferated within themselves through self-symmetry, and the way a tumor grows through intense proliferation of its tissue and vasculature. Treatment methods aside, fractal geometry is useful in approximating the pattern and number of blood vessels present around or in a cancerous tumor, and as such is a good way to track tumor progression. Though fully effective drug delivery to tumors remains elusive, fractal models can in the meantime be used to help physicians in predicting the course of a tumor's growth, and thus in forming more accurate prognoses for the health of patients with cancer.

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Layout by Kara Turner

INTERVIEW WITH PROFESSOR HITOSHI MURAYAMA: SUPERSYMMETRY

By: Sabrina Berger, Junwon Kim, Yana Petri, Kevin Nuckolls



Professor Hitoshi Murayama is the MacAdams Professor of Physics at the University of California, Berkeley. He is also the director of the Kavli Institute for the Physics and Mathematics of the Universe at the University of Tokyo. His research interests include the investigation of dark matter, grand unification, neutrino physics, and physics beyond the standard model, including Supersymmetry.

BSJ: Can you start off by describing your background? How did you get into theoretical particle physics?

Professor Murayama: Well, I was born in Japan, lived in Germany for four years during my childhood, went back to Japan, and eventually got a degree from the University of Tokyo. I found my way to Berkeley as a post-doc up here at the lab, and then acquired a faculty position here.

I don't know exactly the story about getting interested in science. But I was a very curious child, for sure. I was the kind of child who kept asking questions to my parents and so on. My dad was a researcher who worked for Hitachi. He was doing research on semiconductors for the company. He didn't have a PhD, but he had a Masters degree. My memory is, of course, hazy from those days, but I do remember that he answered many of those naïve questions I had at that time, so that's probably how I got interested. That's also how I learned that many questions have answers, which is actually not an obvious thing for many children, I'm afraid. If they're not inquisitive enough, or if their parents or teachers aren't resourceful enough, then many of their questions just go answered. That doesn't nurture curiosity. I was lucky enough to be in that kind of position, I guess.

I was also a very sick child. I had a very bad case of asthma as a child, so I missed many school days. I stayed home quite a bit, so I had to find something to do while I was at home. So, I turned the TV on, the soap operas were not interesting for kids, so I ended up turning my TV to educational channels. Back in those days, in Japan, the educational programs were actually pretty good. Some of them were really sort of story-based. There was one story I particularly remember talking about how infinite series can converge. The story was about a guy in ancient Edo in

“I do remember that he answered many of those naïve questions I had at that time, so that's probably how I got interested. That's also how I learned that many questions have answers, which is actually not an obvious thing for many children, I'm afraid.”

the 17th century who was trying to buy tofu. So, he brings his bowl, and gets one piece of tofu, but wanted some extra. So, he gives many compliments to the tofu shop owner to please him. He keeps praising until, eventually, he got another half piece of tofu. So he continues to praise until the tofu owner gives him half of the rest, and half of the rest, and so on. This guy in the story thought, “Eventually, I'll have a huge amount of tofu, enough tofu for the rest of my life.” But, in the end, he only gets two pieces of tofu. So, that was the story, and it intrigued me.

Another program I remember was a physics program about a little booth, where some man is making some food, and there was a nice aroma. Then comes this strange looking guy who comes close to the booth, smells it, and goes home. He does this everyday, so the owner got fed up with this guy, and eventually gives him an invoice saying, “You've been smelling my food everyday without paying. You owe me 100 dollars.” The rest of the show is spent checking the legitimacy of this request. They first try to figure out what exactly it is that we are smelling. So

they tried blocking the smell off with a thick slab of glass, such that you can see it, but not smell it. This meant that something was actually coming towards you, that could be blocked by the glass. This is how the show unfolds until, eventually, they figure out that there are some particles that come off the food on the grill, propagate through the air, and enter our noses, which explains how we smell things. Of course, I don't know any legal issues regarding the smelling of someone's food.

So these shows were very fascinating to me. I think I was in 2nd grade or something when I saw them, so I talked to my dad and told him "This series thing is kind of interesting." He then bought me a bunch of math books, which I started to read. I studied all the way up to calculus when I was a third grader. I got really, really into it. That's basically how I spent my elementary school days.

For middle school, I moved to Germany, and all of a sudden became very healthy. So, I pretty much lost all interest in those things I had been studying and just wanted to play outside, like soccer and volleyball. I was much more into the outdoors activities now that I could do them. I then got into music, so when I got into college, I became serious about the double bass and got pretty good at it. I was making money off of it as people hired me. Naturally, I considered a career in music until people told me that it was awfully difficult to making a living out of music. But, I always had a sort of interest in physics, remembering those days when I was a kid watching the educational programs on TV. Many of the questions I had asked when I was little were like "Why is the sky blue?" or "Why is it dark at night?" Those questions had to do with physics, astronomy, and some chemistry. Those ideas remained in my mind, so when I got into college, I decided to major in physics. I was studying physics at a minimal level. I wasn't too serious about it, but when I thought about going to graduate school, I thought "Okay, maybe this is the way to go." So, I got into graduate school in physics and wanted to explore the most basic, fundamental thing, which, in my mind at the time, was particle physics. This was because I knew that everything eventually breaks down to tiny pieces, like quarks, atoms, electrons, and other particles. I thought that, in understanding these things, we could maybe understand everything. It was very naïve of me to think this as a senior in college, but that's what I wanted to do.

When I got into graduate school, I wasn't careful enough in choosing the school that was active in my area of interest. I was at the University of Tokyo, so, without thinking, I just applied to University of Tokyo's grad school, which was a really bad idea. No one was active in this area to supervise me, so I was kind of left alone. I ended up seeking people outside the university in the field. Unfortunately, this area was not that active at the time, so not many people were working on it. Eventually,

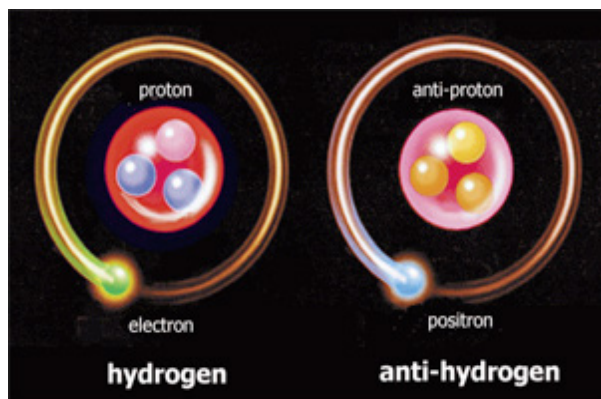
I found somebody at an institute that was 200 miles away. I begged him to please teach me, and finally found him when he was just about to move for a short-term position in England. He said, "Sure, I can do that, but only after I come back from England." So, I lost another two years of grad school. After he came back, I reminded him of his promise, to which he said "I remember I promised, but just teaching one students is a waste of time, so why don't you assemble at least 7 students for me to teach?" I literally went around the country, to cities like Hiroshima and Kyoto, finding students interested in the subject until, eventually, he agreed to teach us.

That was the first time I really got started with learning the subject. The University of Tokyo system is kind of brutal in saying that you have to graduate in the period of five years, no matter what. Given that I only had one year left, it was really tough. I worked incredibly hard, writing a piece of software that's still being used today to compute elementary processes in particle physics. With that, of course, I did a bunch of calculations myself, put together a thesis, and nearly failed. One of the issues with my thesis was that people working on theoretical physics and experimental physics were decoupled from one another. What I was working on was smack in the middle, doing simulations of experiments, but it wasn't very mathematical or theoretical. So, the thesis committee got into a huge debate because they didn't know what side my work would be classified as. In the end, I passed, but was shocked that what I was doing did not seem to be appreciated. I got a degree, so that was great, but it was the message I received that caused me to rethink my career path. That's when I decided to move to another country. So, I applied to the US and was lucky enough to get a post-doc position at LBNL.

Berkeley is really great. After I came here, I kept hearing about the legendary people from Berkeley, one being Louis Alvarez. He is a Nobel Laureate who got the Nobel prize for discovering many particles in the 60's. However, his most famous paper is the theory that dinosaurs became extinct due to an asteroid impact. These subjects don't seem to have anything to do with one another, but that's just Berkeley's style. You just jump in to whatever excites you, disregard what you're an expert in, and just do whatever you want to do. Another Nobel laureate in Physics, George Smoot, got the Nobel Prize by looking at the baby picture of the universe, yet his background also had no implications of this. I was very interested in this intellectual freedom that Berkeley offered, so I really wanted to stay here. I'm really happy about that.

BSJ: How would you explain the basic principles of Supersymmetry to people outside of your field?

Professor Murayama: Well, I think there are two ways of explaining it. One is that it's another version of antimatter. What we've learned really goes back to the 1930's, when every piece of matter or particle we have (electron, proton, quarks) has an antimatter counterpart. Matter and antimatter are actually not very different from each other. So if you just happen to meet a person made of antimatter, you wouldn't recognize it as so. But, the minute you shake hands with that person, you would blow up! That's because when matter and antimatter meet, they annihilate, and turn into a huge amount of energy. So that's antimatter.



So, we now have double the number of particles with matter and antimatter. The idea is to double the number again, and for the same reason. The reason we need to have antimatter is to make the energy of elemental particles fairly stable. Consider an electron. Using, let's say, freshman physics, you have learned about electromagnetism, so you know that if an electron has a negative charge, it repels itself. How do you keep the electron together then? If you think of an electron as a tiny ball with electric charge inside, then you have to put a lot of pressure on that ball to keep it tiny. The amount of pressure that you provide requires energy, which is actually something like at least ten thousand times bigger than the energy that the electron itself has. That's strange,

right? You need to provide that much energy to keep the electron tiny, but ultimately, the electron is far lighter than that. Remember that mass is the same thing as energy, according to Einstein. This was actually a very important puzzle. It turned out that the minute you consider antimatter, there exists another process that will help you squeeze the charge of the electron into a tiny ball, but a lot more easily than before, so that it does not cost so much energy anymore.

Now, you can also apply the same idea to the recently discovered Higgs boson, which was a big deal two years ago. This boson is filling up the entire Universe – it is here, it is densely packed everywhere - and it repels itself, just like the electron does. Then, we're faced with the same problem: if you would like to keep this boson as tiny as it is, then you have to put in huge amounts of energy to squeeze it to be that small, but the Higgs boson does not seem to have this kind of energy. So, where did this energy come from? We use the ideas of Supersymmetry and double the number of particles. Every particle has a partner, each with a strange name: the photon has a partner called the photino, the gluon has a partner called the gluino, and the electron has a partner called the selectron. Once you accept this extra partner, then the previously unaccountable energies become okay again. So that's one explanation.

The other one is the idea of extra dimensions. So, we live in three-dimensional space, but there are ideas that our space is not actually three-dimensional, but maybe nine-dimensional. This is what string theorists tell us. The extra six dimensions are curled up in a very tiny size so that we don't actually see them, but they do exist. At least, that's the idea.

In Supersymmetry, we have yet another type of extra dimensions. Ordinary extra dimensions, even though that already sounds extraordinary enough, can be described by numbers. Think of an intersection. We can decide to meet at 5th Avenue and 3rd Street on the 7th floor of a building. This is all described in a coordinate system,

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Standard particles					Supersymmetry particles				
u	c	t	g		$\bar{u}$	$\bar{c}$	$\bar{t}$	$\tilde{g}$	
Up	Charm	Top	Gluon					Gluino	
d	s	b	γ		$\bar{d}$	$\bar{s}$	$\bar{b}$	$\tilde{\gamma}$	
Down	Strange	Bottom	Photon					Photino	
ν_e	ν_μ	ν_τ	Z	H	$\bar{\nu}_e$	$\bar{\nu}_\mu$	$\bar{\nu}_\tau$	$\tilde{Z}$	$\tilde{H}$
Electron neutrino	Muon neutrino	Tau neutrino	Z boson					Zino	Higgsino
e	μ	τ	W		$\bar{e}$	$\bar{\mu}$	$\bar{\tau}$	$\tilde{W}$	
Electron	Muon	Tau	W boson					Wino	
Quarks Leptons Force particles					Squarks Sleptons Neutralinos & Charginos				

described by a set of numbers. But, Supersymmetry is an extra dimension, whose coordinates are numbers, but they don't commute with each other. When you have two numbers and change their order, you get an extra minus sign. It's a weird kind of number. It's a dimension nonetheless. So, another way to describe Supersymmetry is a structure of new dimensions in space. When a particle goes into that weird dimension, it comes back as a partner. An electron enters, a selectron exits. A photon enters, a photino exits. So, that's the results of this new dimension of space. It's a quantum dimension of space.

BSJ: How or why did you choose to pursue Supersymmetry over other models in particle physics?

Professor Murayama: I didn't necessarily choose it over the other theories out there; it just seems to be the best understood and the most viable. Mathematically, Supersymmetry is quite beautiful. Its ideas have been useful in many developments in recent mathematics, like in understanding topology of four-dimensional spaces, or even six-dimensional spaces. It turns out that Supersymmetry has a very rich mathematical structure. It seems to help with the idea of why particles don't have as much energy as we think they should. It also helps us understand an even bigger version of the Supersymmetry's grand implications.

We have identified at least four different forces in Nature – electromagnetism, strong force, weak force, and gravity. There is a possibility that all of these forces actually come from a single force at the beginning of the Universe. They may have separated as time went on, progressing to the way we see four different forces now. That idea also requires Supersymmetry to make it consistent with the data we currently have. It also provides a candidate for something called the dark matter theory. Dark matter is everywhere and is, in some sense, the mother of stars and galaxies, which were made thanks to dark matter. When the Universe started, it was a very bland, boring place, totally smooth and looked exactly the same everywhere. But, eventually, the Universe managed to create these bumpy structures that are stars and galaxies, or else we would not be here. So, how did the Universe become so bumpy? For the answer to this question, we turn to dark matter. We can't see this dark matter, but we know where it is, we know it exists. Dark matter has enough gravitational pull to assemble things together just by pulling them using the gravitational force. Where dark matter is a little bit more dense, it pulls stuff in to become more dense. As the gravitational force becomes stronger, it can pull in even more, and become even denser. Eventually, the universe forms these clumps of densely-packed dark matter. Those clumps draw ordinary atoms in, which scatter against each other, emit light, cool down, and eventually collapse into stars and galaxies. That's the best theory we have.



gravity



strong force



weak force



electromagnetism

Unfortunately, we still don't know what it is yet. However, Supersymmetry gives us a candidate of this dark matter particle, so it is also quite useful in that way.

BSJ: In several of your papers, we read about the hierarchy problem. We wanted to ask you about the ways Supersymmetry can attempt to resolve this problem.

Professor Murayama: This relates back to what I mentioned about the energy of the particles. The hierarchy problem is the problem that the Higgs boson, which we now know weighs under 25 GeV, or gigaelectron volts, could have also been at the highest scale possible, 10¹⁸ GeV, which is the highest energy scale we could ever imagine. Because we know that the Higgs boson is much lighter, so something must be protecting it. Initially, the idea with the electron was that something is indeed protecting it, so it can be much lighter than it would be alone. But, because of the presence of antimatter, with antimatter actually cancelling part of this dark energy, the electron can remain light. In the case of Higgs boson, again, if you consider it alone, it's mass tends towards the highest possible energy scale, so we know something should be protecting it from remaining at this huge mass and energy. Something must be cancelling its self-repelling force, so we use the ideas of Supersymmetry. That's how the Higgs boson can stay as light as we have discovered, we think. This is one of the greatest influences Supersymmetry has in helping us understand these tiny particles.

BSJ: What is the relationship between Supersymmetry and String Theory? How can they be used to further our understanding of dark matter?

Professor Murayama: String Theory is a theory that all particles we see are actually not points, despite what we used to think. They are actually these extended rubber bands, which have many branches. Obviously, these rubber bands have to be small enough so that we can perceive them as points. So, they are tiny, tiny strings,

“If you create a universe where these fundamental constants are just a tiny bit different from what we have in this universe, would that result in a totally different universe? Would there be life? Would there be people?”

but are extended and have a finite size. People think that this might really help us unify gravity with the other forces that we know in nature, so that they ultimately come from a single force, which I mentioned earlier. So, why does this extra size help? Well, think back to the very beginning of the Universe. We know that the Universe was getting bigger and speeding up as it did so. Thus, the Universe should have been much smaller before. If you just keep going further back in time, eventually, the universe reduces to a single point. Here's where we don't know what is going on, because the entire energy of the Universe collapses to a point, the density inside this point is infinite. Whenever physicists see infinity, we throw up all hands, for we don't know what to do with it. The laws of physics as we know them just break down. We struggle to study the question of how exactly the Universe got started because we don't know how to handle this infinity.

But, just imagine that the Universe was, instead, made of these tiny strings, not elementary particles. Then, as we try to squeeze the Universe down to a point, it gets stuck, because the strings inside have finite size. This gives us hope that once we understand String Theory, we can successfully avoid this infinity. We can probably understand how the Universe got started without getting into this issue of infinities, which is the way that String Theory helps resolve this problem. When people started to build String Theory, they quickly discover that we would need the ideas of Supersymmetry in addition to the tiny string to make the theory mathematically consistent. So, that's where the idea came from. Bruno Zumino, from our department, who unfortunately passed away last year, and Julius Wess, who also passed away a couple of years ago in Germany, were the people who wanted to implement the idea of Supersymmetry in a kind of theory we could deal with, calculate with, and use to make predictions. Since 1994, the idea of Supersymmetry really took off as it was combined with everything we now understand about particle physics. We now have this new theory, with which we can make predictions about what kind of signals we are supposed to see at the Large Hadron collider, some

experiments in cosmology, cosmic ray experiments, and so on. This is the frame that can combine many other theories together into a single theory. Of course, we have not seen evidence of it yet; it remains undiscovered, but at least it is something we can think about, deal with, look for, and study.

In connection with dark matter, we've mentioned that Supersymmetry predicts the partners for every single particle we have in the Standard Model. There is a good reason to think that photons is one (among the whole host of supersymmetric particles) that is stable, does not decay, is electrically neutral, and weakly interacting. It is actually one of the best candidates for dark matter particles. There are people in this department who have pioneered an experiment done underground, looking for a signal for dark matter. Of course, again, we have not found it yet, but we are getting into the range of precision where you might expect a signal, so that's quite exciting.

BSJ: Also, in one of your papers, you mentioned the conflict between naturalness and Supersymmetry. Could you first explain this philosophy of naturalness, and then explain some of the conflicting concepts we see between Supersymmetry and naturalness?

Professor Murayama: Naturalness is the idea that we have this Universe with lots of physics. Most of physics has many intrinsic numbers in it: the speed of light, the mass of electron, electric charges, the mass of the proton, strength of the weak interaction. We describe physics using what we call the fundamental constants. Suppose, now, that you play the role of God and are thinking about making a universe. You have to choose these fundamental constants to set up a universe, but you don't have any particular reasons to choose one number over the other. If you create a universe where these fundamental constants are just a tiny bit different from what we have in this universe, would that result in a totally different universe? Would there be life? Would there be people? Would there be stars? When this other universe, which has slightly different fundamental constants, looks pretty much the same as ours, then that gives you some sense of stability in our universe. This universe is sort of “natural” in relation to the stability we see. If you tweak things around a little bit and yield a totally different universe, then we think this kind of universe is “unnatural”, because you need to choose these constants extremely precisely so that we can live in this universe and have stars and galaxies and so on so. That's the concept of naturalness.

Of course, naturalness is not completely scientific because we can probably never observe if there are other universes at all, but maybe they don't exist. It may just be totally ludicrous to think about changing these fundamental constants around. Maybe there is a way to

actually derive these fundamental constants from some principles. Maybe they are supposed to be exactly the way they are. There's always a philosophical debate about this subject.

This problem I mentioned about the mass of the electron or the mass of the Higgs boson relates nicely to this idea. If you change things by just a tiny bit, at the order of 10^{-36} , the Higgs boson would be much more massive. It doesn't seem to be natural in that sense. So, here lies our problem.

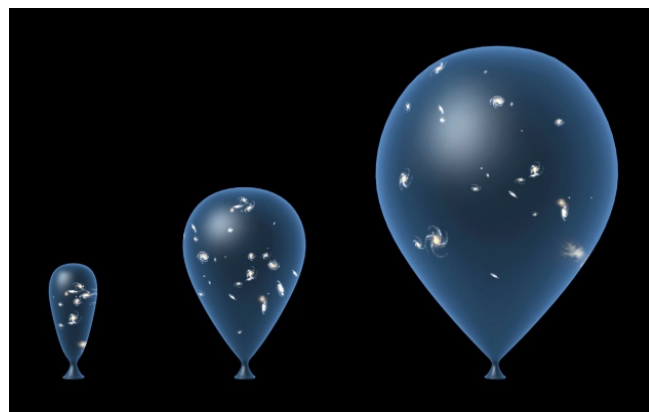
BSJ: In trying to test these notions of naturalness and other constants, do theorists just take current constants that they know, tweak them slightly, and just see how other physical processes work out?

Professor Murayama: Yes, exactly. In doing so, though, we see that most things are not that sensitive to change. For example, if you change the mass of the electron, say make it twice as big, atoms become twice as small. This doesn't really seem to change things very much. In contrast, the difference between the mass of the proton and neutron is more sensitive. The mass difference between the proton and the neutron is only about 0.15%, so they are extremely close in mass. If you make the neutron 2% heavier than it is now, then all the neutrons in your body very quickly decay into protons which are lighter. Then, because the neutrons act as the glue for binding the nucleus together, these nuclei can no longer stay together. Thus, the protons (again like charges repel) would all of sudden blow apart, so you wouldn't exist. If it's the other way around, in which the proton is less than 2 % heavier than the neutron, then protons decay into the neutrons, causing nuclei to become electrically neutral.

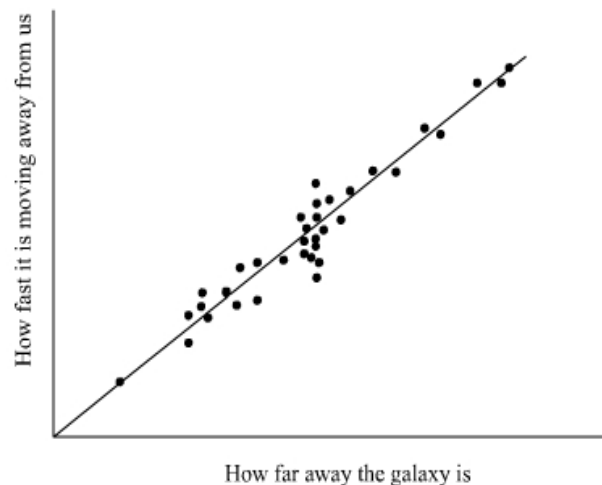
Then, there would not be any atoms. There would not be any periodic table or chemistry. There wouldn't be any humans. This case is a little more sensitive than the case of the mass of the electron, but again only at the level of a few percent.

There are two things that seem to be, in this sense, very unnatural. One of them is the Higgs boson. If you change things around just a tiny bit, at the level of 10^{-36} , as I previously mentioned, the Higgs boson becomes enormously massive. It would then be stuck in the universe today, unable to move. Then, all the elementary particles would become massless, electrons in your body would fly away with the speed of light, and you'd disappear in a nanosecond. So, this tiny change would wreak such havoc in the universe, thus making this very unnatural. That's one example.

There's one other example where you are incredibly sensitive to these kind of small numbers, and that's the current acceleration of the universe. We still don't know exactly why the universe is picking up speed these days. Saul Perlmutter discovered this and got a Nobel Prize



for it. It's named dark energy, which is filling the entire universe. This sort of sounds similar to the Higgs boson, and they must be related at some level. This dark energy is multiplicative in nature. If you make the universe twice as thick, the volume of the universe becomes eight times bigger and dark energy becomes eight times bigger. It keeps pushing the expansion of the universe as it gets bigger, so we see the universe accelerating at an increasing rate. Now, suppose this is true and somehow empty space has this dark energy. Because it grows with volume, there must be some constant density of energy in the empty space. But, who chose this constant? Again, I can play the role of God here, where I change this number a tiny bit and see what happens. It turns out that if I change things around only the tiniest bit, even worse than the Higgs boson, at the level of 10^{-120} , this energy density of the universe can become hugely positive or hugely negative. If it's hugely positive, then the universe must have expanded or started when it was still very hot and dense. Then, as things start to accelerate right away, everything splits apart and there's no time left for stars and galaxies to form. On the other hand, when it gets driven to this huge negative number, as the universe gets started it actually decelerates so quickly because dark energy is negative. It stops right away, starts to collapse, and leaves no time for any stars or galaxies to form. The way the universe is today seems to be very sensitive to this dark energy, or whatever it is that decides this energy density of the empty space.



“...the way it works is that experimentalists tend to be glued inside their laboratories. We tend to be glued at our desks, working on our computers and stuff. Thus, it is actually not easy for us to meet and work together. We actually have to make a conscious effort to do so, so that we can contribute to each other’s work.”

If you require that the amount of dark energy has to be as such to be able to build galaxies and stars eventually, then you can predict that the amount of dark energy today must be within about a fraction of 10 of what’s all discovered. The number we got in the end seems just right. If we change things just a tiny bit, we wouldn’t be here, so we say again that it seems very unnatural. From what I know, these two numbers are the only numbers in physics I know of that seem so sensitive to tiny variations and seems so unnatural to us.

BSJ: When pursuing your research in your career in particle theory, a lot of your work is highly collaborative with the high energy experimentalists. I was wondering if you could explain how your relationship with various experimentalists, especially those working at the LHC, affects how you choose the work you work on?

Professor Murayama: Experimentalists are, first of all, very important. I’m kind of jealous because of what the experimentalists’ ability to talk to mother nature. Theorists are sort of receiving second-hand information. We ask the experimentalists to do the experiments. They know how to talk to mother nature, so they get some answers. They then actually consult us, theorists, again and they say, “We got these answers, but they’re so cryptic, so we can’t make sense out of them. What do you think of mother nature’s response?” That’s where we come back in. I really admire them. We really need them so that we can get information about the way the universe works in the end. I very much love the idea of collaborating with them.

We, the theorists, start by giving them advice or suggestions about interesting directions to take in their work. Then, the experimentalists go ahead and build some complicated instruments to take data. Then, they bring the

data back to the theorists and say, “You know I tried what you suggested, we got this answer, so what does it mean?” Then, we start the process over again. So, that’s the way that science is supposed to make progress. I try to remain very close to them in my work. In practice, the way it works is that experimentalists tend to be glued inside their laboratories. We tend to be glued at our desks, working on our computers and stuff. Thus, it is actually not easy for us to meet and work together. We actually have to make a conscious effort to do so, so that we can contribute to each other’s work. I am honored to be invited to many advisory committees and make suggestions on how big laboratories should be run, how the next experiments should be chosen, and what is the right way to take data. I contribute my two cents; and sometimes they listen, sometimes they don’t. That’s okay. That’s what really sets physics apart from philosophy. Physics is really based on the data. I have nothing negative to say about philosophy by the way, but that’s the difference.

BSJ: Along the same line of thought, just to clarify, although you work together, theoretical and experimental physicists have very different ways of approaching the same problem. How would you break down your abstract ideas to establish testable hypotheses that the experimentalists can work with?

Professor Murayama: One of them is to basically convert ideas to numbers. For example, let’s say I have my own idea of what dark matter may be. Then, that idea itself is difficult to test. If I want my idea to be tested, then I have to come up with a set of data that could be experimentally obtained that should match that produced by my new theory. Once I’ve made this definite prediction, then experimentalists can work from their side of the problem. They can start to think about exactly how they can build an instrument to be able to take such data, and make sure that the instrument they build is sensitive enough to be able to accurately agree with the numbers I’ve come up with. Once the problem is concrete, then they are really the experts in doing work on this problem. I just need to make sure to make my abstract-sounding ideas as concrete as possible. Therefore, in the end, I just predict a couple of numbers that are supposed to come out from a very particular type of experiment and see if they agree with the numbers I predicted. Although this process typically takes a long time, I think it’s the only way we can work together.

BSJ: Now, when you’re coming up with these concrete numbers, do you present experimentalists with maybe a list and see what might be easiest to test?

Professor Murayama: Yes, absolutely. A list of numbers, or some plots or programs that they can also

play with and see what numbers will come out that might be easiest for them to test.

BSJ: We read in one of your papers that you mentioned X-ray observations of galaxy clusters could provide support for the dark matter hypothesis. What other evidence could be used in support of supersymmetry?

Professor Murayama: Many things, in fact. For example, if dark matter is really made of this supersymmetric particle, it's supposed to fill the inside of our galaxy. These particles are very shy, and don't interact at all with matter most of the time. This is why we don't feel the wind of dark matter all the time. But, once in a while, these particles may decide to annihilate with each other, just like matter and antimatter do, and produce something that we can observe. It might be a very energetic photon, in the form of gamma rays, or maybe particle and antiparticle pairs. Then, these made particles may eventually propagate over the disk of a galaxy and fall from the sky so that we can observe them. This idea, actually, has been discussed quite a bit in the last several years. For example, with gamma rays, which again are very energetic photons which may come from this dark matter, there seems to be an indication that there are actually quite a few of these photons coming from our galactic center, which is presumably where the dark matter is most concentrated within a galaxy. So, that raises some hope.

Also, once antimatter is within a galaxy, and when it takes a sort of "random walk" through this galaxy, no one expects that this antimatter would meet up with a matter particle and annihilate. But, if you do see antimatter particles coming from space, then you should wonder where they're coming from. They couldn't have travelled very far, or else they would have all annihilated away. They must have come from fairly nearby, which, from a galactic standpoint is about a hundred thousand light-years. But,

regardless, this is relatively nearby. So, we need to look for what produced these antimatter particles. There are locations in the galaxy where very energetic particles are produced, like supernova remnants. This is a, sort of, nearly dead star that had exploded at the end of its lifetime and has left these supernova remnants that are still spurring out from its core, which is slowly sizzling and fizzling out until it fully dies. But, before it is totally dead, it still spurs out these energetic particles. So, maybe, that's the location we're looking for. But, we can spot some of these, and if they're not coming from the right direction, or if they don't seem to be producing enough of them, maybe this could also be evidence that these dark matter particles inside a galaxy fairly nearby might have annihilated with each other and produced a pair of matter and antimatter. Then, that antimatter managed to survive and reach us. So, that's another piece of evidence that other people are looking for.

BSJ: This is more on a tangent to what we've been talking about until now. Many people in the field of mathematics or the field of theoretical physics who have been in the field for a long time build a rather intangible sense of intuition. This intuition allows these people to essentially look at a problem that they'd like to answer and gauge not only the difficulty of a problem, but also the complexity of the solution it might yield. So, we were wondering if you might be able to talk a little bit about how you first interact with a problem and determine its solvability and how you approach an initial solution.

Professor Murayama: Well, my approach is pretty simple. I will usually work on a problem for a while until I hit a brick wall. Then, I just, well, leave it and start working on something else. Then, if I hit a brick wall again, then I leave it and start something else. I end up taking this random walk between problems. Now, once in a while,



A ghostly ring of dark matter.

I manage to break through this brick wall and come up with an answer. That's how I make progress.

Then, I remember the problems that I've left behind and decide to spend a little more time on one again. So, I go back to that one and, surprisingly, even though I had been doing something totally different for a while, somehow, my brain matures and improves. I can then, sometimes, get through the brick wall I had hit before. I'm actually not a very patient guy, so I'm not very persistent in really working hard on a particular problem for years and years. So, I tend to jump around, which has served me pretty well because the field also has many directions and is quite fluid in directions others may be taking. So, I'm okay with this.

Of course, in the end, I'd like to solve really, really hard problems. So, for some reason, jumping around and talking to many people really helps. Berkeley's a great place with so many wonderful people. I'm not patient enough to read every single paper that appears in arXiv, every single textbook on the matter, but talking to people seems to allow me to learn things much more quickly. There really are so many people to talk to in Berkeley, so that's the way that I end up learning and making breakthroughs. I don't recommend this to students. You are supposed to solve your homework problems, but that's my style.

BSJ: Now, before you hit that first brick wall, though, many of these problems are quite abstract and hold many different approaches towards a possible solution. So, how do you first come to an initial path to trying to solve some of these problems?

Professor Murayama: Well, that's an interesting question. That, of course, is determined on a case-by-case basis. So, I would say that if the problem is already familiar enough to me, meaning that I already have enough information in my brain to think on my own and try to stitch together these pieces of information to come up with a solution to a problem, then I tend to spend only a few days or a couple of weeks on the problem. Then, I can sometimes see if I'm making any progress. If not, then I stop. If the problem I want to work on is sufficiently unfamiliar to me, then I have to start reading materials on the subject and familiarize myself with the problem. This includes learning about the many techniques other people have used to solve similar problems. Then, I start talking to people and attending lectures. That may take a few months. Then, after talking to many people, I will at least get some sense of what has been done already and what is still a big problem. Then, I have to eventually decide whether or not I want to pursue the problem. So, it varies.

BSJ: So, we have read about your position as the Director of the Kavli Institute for the Physics and Mathematics of the Universe. Could you comment on the philosophy

“Berkeley's a great place with so many wonderful people. I'm not patient enough to read every single paper that appears in arXiv, every single textbook on the matter, but talking to people seems to allow me to learn things much more quickly.”

behind such an institution and the role it plays in your research field?

Professor Murayama: So, there are a lot of things in common between Berkeley and that institute. It's just a different organization. So, when I founded this institute, I had this idea that, if we could just break the walls down between different departments and disciplines, what are some helpful combinations of disciplines that would allow for us to make the most efficient progress? What I saw was, especially for people working in string theory, a much more advanced mathematical theory of physics, they interact with mathematicians. They have to because they need advanced mathematics. Mathematicians also want to learn from the string theorists to gain inspiration for some of their work. So, that actually works out to be a very good combination to have.

The kind of thing that I do is much more experiment-oriented, so I love to have experimentalists nearby. A lot of other things that I have a lot of connections to astrophysics and astronomy, so it's also good to have them around. So, in the end, the idea was to have this collection of disciplines meet, which normally, in a university setting such as Berkeley, are divided in different buildings. If you have an institute where everyone is together, everyone sees each other everyday, then you tend to come across some breakthroughs that would not have been discovered by these people, individually. So that's the way this institute was designed.

It's a different structure. Being in a traditional department of course has some advantages by allowing people to pursue fields at a deeper level within a discipline. You have enough expertise from different people within the same area, so it's easy to talk with each other. It's also a lot easier to train students if you're within a particular discipline, for once you go outside the border of a discipline and want to be more interdisciplinary, it's hard to figure out what degrees students would earn. That sort of mundane issue is also important. It's also had to figure out what journal they should publish in. It really isn't clear. So, for students, it's probably a lot more comfortable and

easier to be in this structure containing more traditional departments because everything else is currently structured that way. So, there are pros and cons. What I hope for is that when I'm over there at the institute, I interact with people from different areas and, as a result, I actually became in charge of building an new telescope, which I've never done before. I don't think that I'm good at it, but I can still organize the group of collaborators that will be working on this. It's an \$80 million project, for which I had to raise funds, which I'm also pretty good at. So, I can play my role towards a very different goal from what I used to do. So, that was an opportunity that I don't think I would have ever had if I were just a physics professor in the physics department here at Berkeley. It comes with this extra cost, in that I should spend some extra time and learn how to talk to people in different disciplines, which can sometime be a bit confusing. You'd be amazed, once you get more specialized and go to graduate school for a particular discipline, that it becomes much more difficult to talk to people from other disciplines because every discipline cares about how precisely you make statements. The precision means different things in different fields. The word we use to describe precision is different from one field to another. So, just communicating is rather challenging. It's like someone who speaks French and someone who speaks Chinese trying to talk to one another. Theoretical physicists talking to mathematicians is like that, actually. We speak very different languages.

to solve. So, that's the way I choose to grow my horizon. Sometimes, I hit a jackpot. I try to pursue it a bit further until I hit another brick wall. This is another way to grow in my research and has been the way that I have done so. I don't really know what I'm going to be doing, but at least I see some particular directions that seem rather fruitful, which I hadn't imagined before, but seems to be coming out very nicely.

BSJ: Thank you very much for your time.

Professor Murayama: No problem. Thanks for having me.

IMAGE SOURCES

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Layout by Abigail Landers

BSJ: You were talking about constructing this telescope in the future, but what are some future steps and directions you plan to take with your research?

Professor Murayama: Let's see. So, I've been relatively random in what I do; I jump around. I also participate some underground experiments studying neutrinos. Unfortunately, that particular experiment was not one of the ones that got a Nobel Prize this year, which I believe it deserved, but it didn't.

As I said, I tend to be relatively random, so I don't really know. But, I can imagine that, now that I'm trying to build this instrument for the telescope, I'm sure I'd like to use it to take data of my own and analyze it. So, that's one direction I could certainly imagine. Some other things I've been doing with postdocs and students have gone off in very different directions in mathematics, which I was not familiar with. I used to use a lot more geometrical techniques, but these new techniques tend to be much more algebraic. I knew very little about it when I started, but now I know quite a bit. We believe we actually made a very important breakthrough just yesterday, so I'm very happy about this. Well, certainly, this new technique I just learned seems to be very versatile and should be applicable to more problems than the problem we have just managed

AN INTERVIEW WITH PROFESSOR GIAN GARRIGA ON ASYMMETRIC CELL DIVISIONS: DISTINCT FATES OF DAUGHTER CELLS

Manraj Gill, Tiffany Nguyen, Georgia Kirn



Figure #1. Professor Gian Garriga

Dr. Gian Garriga is a Professor of Genetics, Genomics and Development in the department of Molecular and Cell Biology at the University of California, Berkeley. Professor Garriga's interest in understanding the *C. elegans* nervous system has led to a study into more fundamental questions of cell biology. In this interview, we talk about one such topic, asymmetric cell division, and discuss not only its molecular basis but also its role in apoptosis and stem cell differentiation.

Berkeley Scientific Journal: How did you first get involved in your field of research?

Gian Garriga: After college I was pretty sick of school so I actually did other things for many years and then sort of accidentally met some people and ended up going to graduate school. I was a molecular biologist and a biochemist. As a graduate student, I studied RNA splicing. At the end of that, the original plan was to go and get a job in the industry but I thought, "Well, I could put that off."

I looked around for things to do as a post-doc and thought that it would be good fun to work on something that really wasn't understood at all. At the time, people didn't really know how the nervous system was developed. It was very different from what I had done previously so I looked at different organisms where people were studying this. And even though I didn't do genetics as a graduate student, I sort of appreciated it.

It was really a choice between doing something in *Drosophila* or *C. elegans* (*Caenorhabditis elegans*). *C. elegans* were this sort of newer organism in the sense that people had only recently been studying it. And it was also very simple and people appreciated that and you could freeze it. [That's something] you couldn't do with flies and I'm not that organized so something I could freeze was better.

BSJ: And what led to this focus into cell division?

GG: That began just when I started working as a post-doc, I worked on a pair of motor neurons that innervated egg-laying muscles and stimulated hermaphrodites to *C. elegans*. They stimulated hermaphrodites to lay eggs. So, I started to

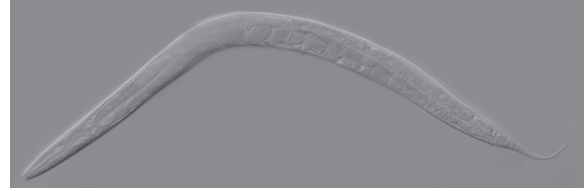


Figure #2. *Caenorhabditis elegans*

screen for mutants that were defective in various aspects of the development of these neurons.

We identified genes that were involved in all kinds of aspects: from how the neurons were generated and how cells migrated early in their development to how they later send out axons. Just because of the genes that I found most interesting when I came to Berkeley, I focused on asymmetric cell division.

Migrating cells would polarize growth cones, which are the ends of axons that are migrating. And then these cells that divide asymmetrically give rise to different types of cells during development. It's sort of a polarity issue. Polarity drives all of these processes, so that's what we've been working on pretty much since I've been here. Which is a long time...

BSJ: Why study *C. elegans* specifically for this question of cell division?

GG: *C. elegans* is the only animal where we know its development in detail. People in the 1970's and 1980's began to just follow the divisions of *C. elegans* because it's simple and transparent. You didn't need to have any sort of special methods. And you could observe the cells divide.

John Salton, who won a Nobel Prize for this, was able to follow all of the divisions in *C. elegans*. They're stereotyped between one organism and the next. The lineage is invariant and you can predict where any cell has been, in terms of its ancestry. You know if it's a precursor cell, it's going to divide. And if it's not, you can tell what type of cell it is: if it's a neuron, if it's a muscle cell, or if it's a cell that's going to die.

One of the things we study is apoptosis. To understand how fate is specified, and understand the process of asymmetric cell division, you really have to understand this lineage and how that's generated.

BSJ: What exactly is asymmetric cell division?

GG: You can think of how you specify cells in two different ways. I'm going to pick flies as examples here. When I first came to Berkeley, there was a lot of work in Jerry Rubin's lab on fly eye development. The way that works is there's an undifferentiated epithelium as there's a morphogen that

sweeps through. And in its wake, you start to assemble these structures called ommatidia. The eye of the fly is this repeating unit of ommatidias that have a number of photoreceptor cells and support cells. You start to specify individual cells. The first one is the photoreceptor cell called R8. Through interactions of R8 and the surrounding epithelial cells, the fates of the other cells are determined. You generate these ommatidial units. If you look at the lineage that gives rise to that, any cell can be related to any other kind of cell, so there's no lineage at all.

If you look in flies at other structures, sensory organs within the fly, they're generated by a particular lineage. So, a cell would divide and it will assign distinct fates each time to the two daughter cells and those divide to generate two daughter cells that have distinct fates. That process where a cell would divide to generate daughter cells that have distinct fates is asymmetric cell division.

Stem cell divisions are like that too. With a stem cell lineage, it'll generate another stem cell but also a cell that's more limited in its developmental potential. That would be considered an asymmetric cell division.

So, basically any division where you give rise to two daughter cells that have distinct fates.

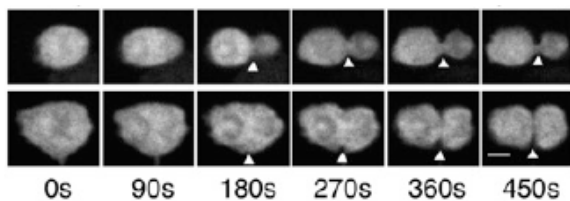


Figure #3. One type of asymmetric cell division (top

BSJ: There's this idea that specific proteins being segregated differently would lead to these distinct fates. What exactly causes that distribution?

GG: Some molecules have been identified, in *Drosophila* and in *C. elegans*, as being involved in asymmetric division and encode proteins that are localized asymmetrically during the divisions. The molecules are conserved and they contain similar roles in all kinds of animals, including us. How they get distributed can really vary.

In some cases, you inherit the polarity from the cell from which it's coming. An example of that would be the *Drosophila* neuroblast; these are cells that will divide to generate the neurons in *Drosophila*. They come from an epithelium that's polarized and they inherit aspects of that polarity from the epithelial that they were originally. They delaminate from this epithelial layer, but they inherit the polarity of those epithelial cells. The polarity is subsequently used to generate asymmetries in division.

In other cases, the cells are polarized by cues and signals from the environment. Those signals polarize the cell and they then divide asymmetrically.

BSJ: What exactly causes the axis to align perpendicularly

to where the epithelial is or wherever the separation is?

GG: If you have a cell that's dividing and you have something that is asymmetrically distributed on one side and if you want that to be inherited by one of the cells so it will adopt the fate different from its sister cell, then it's really important to align the spindle in a particular way. If you align it [parallel to the separation], both of the cells are going to inherit that asymmetrically divided protein and generate the same fate.

There's a hierarchy of molecules that are involved in this. Those molecules at the very top of the hierarchy are involved in not only controlling the distribution of fate determinants, proteins or RNA molecules, but also in controlling the orientation of the spindles so that the cell divides properly.

BSJ: Are there different types of asymmetric cell divisions?

GG: Yes! There are cell types that divide asymmetrically that are controlled by intrinsic polarity of the cell itself and there are divisions where the polarity is imparted by signaling molecules. You can even get cases where a cell divides and there is an inherent difference in the cells. They're played out by interactions either between the cell types or between the cells and the environment.

You can imagine a case where a protein is inherited by one cell type. An example of this is the Numb protein. The fate of the cells is determined by Notch signaling, but Numb interferes with notch signaling in one of the daughter cells. That Notch signal can come from the environment or sometimes even just from signaling between the two cells. So, there's a bias in the notch signaling.

The other [kind of thing that can happen is controlling the spindle by controlling the position] of the spindle where the cells that are generated are different in size. In *C. elegans*, we think it plays an important role in apoptosis. We don't know why but there's a good correlation between mutants that we've identified over the years that affect [the ability of the cell to survive. Some die and the cells that normally would die survive.] Divisions that are highly asymmetric, generating a much smaller cell and a much larger cell, generate cells that are going to die. The larger cell survives and the smaller cell dies. In mutants where the cells actually survive, [the position of the spindle is affected. Or at the very least, the division plane is affected.] The more symmetric the division is, the more likely it is that the cell survives.

BSJ: Is the smaller cell always fated to die?

GG: There are lots of divisions where you generate cells in different sizes where the [smaller] cells don't die. There's something different about this lineage but there's some aspect of the asymmetry of the division that is contributing to the apoptotic fate. What this is we don't know. One way of thinking about it is that you distribute molecules in the cell so that when one cell divides the smaller cell is not going to have enough something to allow it to live. And if you now make the division more symmetric it may get more molecules that control the ability to survive. So, there's a good correlation between cell size and ability to survive in the lineages where

the cells normally would die.

BSJ: In terms of apoptosis, are the triggers coming from the parent cell, maybe some apoptotic factor? Or is it that once a daughter cell doesn't have whatever resources it needs, it's conditions triggers apoptosis?

GG: We don't know! All we do know is this correlation and we don't understand this mechanism that drives that correlation. Apoptosis in *C. elegans* is pretty well understood so we really understand how cells execute this [conserved] apoptotic pathway. We know in certain cases that the apoptotic fate is controlled by transcriptional regulation of the most upstream apoptotic gene. So, if you turn that on, you kill the cell.

But there's this additional thing going on where there has to be a control of the positioning of the furrow and that contributes to the apoptotic fate. Normally the furrow would be in the middle of the cell. But these cells go through the effort to displace this furrow. Most of the molecules that we're studying that we originally thought were involved with apoptosis, we think are really involved in controlling asymmetry of division in terms of furrow position.

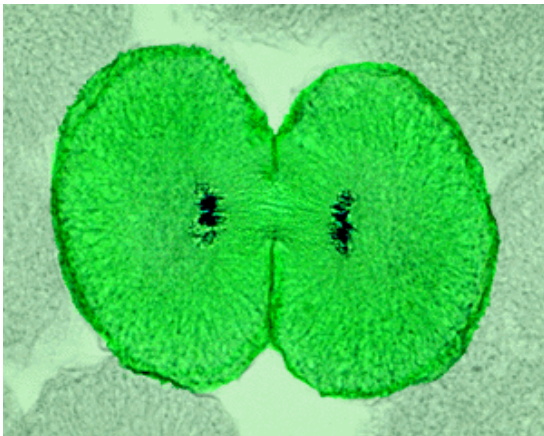


Figure #4. A cleavage furrow in the middle of the cell

BSJ: What are aggresomes and how does asymmetric inheritance of those influence apoptosis?

GG: Nobody knows why cells die in *C. elegans*. In some other cases it's pretty clear why you would kill a cell... During limb development in mammals, the cells between the digits die and if that fails [to happen], you get fused digits. There are places in our brain that 50% of cells generated die and that's a little less clear why you would do that. In *C. elegans* there's a few cases where you would understand why cells would die! There are some cells that are sexually dimorphic, so they're used in the hermaphrodite and killed off in the male or vice versa. The other places where you have these lineages that are repeated on the anterior or posterior axis and you may need some cells near the middle of the animal but not outside so you'd kill those off. But for the most part in a hermaphrodite there are 131 cells that die and we don't know why.

So, there's been speculation on one idea that these

were pseudo genes, which are thought to have no function and are found all over the place in genomes. So this idea was called pseudo-cell hypothesis. It's kind of an evolutionary argument: if you allow a cell to die, it can drift in function and if you allow it to survive most of the time, it won't have a new function and will be the equivalent of a pseudo gene or pseudo cell. But in some cases it may acquire new functions that would have adaptive value.

We were asked to review some papers that we thought were interesting and might be related to apoptosis. We came across this paper in *PloS Biology* where they were overexpressing this Huntington protein with amino acid repeats that cause abnormal folding and cause aggregates. They were just expressing these aggregates in tissue culture cells which people have done and they saw that these Huntington protein aggregates would sort-of overwhelm the ability of the proteasome to degrade it. And when you do that, there's this mechanism where they form these aggresomes. [Aggresomes] are then transported back along the microtubules to the microtubule-organizing center are dealt with there. This paper went one step further and watched what happened when cells divide. So you duplicate those organizing centers to generate centrosomes and they found that one of the cells always inherited the aggresome asymmetrically. They went on to express these in *Drosophila* neuroblasts that generate asymmetric division, and they found they were asymmetrically distributed to the stem cell that actually died before the neurons.

So, the idea was that this was a mechanism to put these aggregates of proteins into the longest living cells. They even looked at the intestines of people with a neurodegenerative disease. In intestines, there are these asymmetric stem cell divisions and there they found, even with no pathology there, these cells divide and generate another stem cell and then a cell with more limited developmental potential. They saw that [the cells with limited developmental potential inherited the aggresomes].

So, the idea is that you protect the cells from these aggregated proteins by distributing them to the cells that'll be around less. We saw that and proposed that maybe that's what's going on in *C. elegans*: the cells that die are trashcans. I still think that's a really good idea but I haven't convinced anyone to test this by misexpressing these proteins high enough to produce aggresomes; this is just an idea.

BSJ: Which comes first... this distribution of aggresomes that then triggers apoptosis or apoptotic signals that then attract the aggresomes?

GG: Right, I would predict that you would generate aggregates of proteins under certain situations that are bad for the cells and that something evolved to get rid of that. But there are plenty of places that are important developmentally to have apoptosis so apoptosis could've evolved independently and then gotten used [for these other purposes].

BSJ: How do you approach these questions and what methods do you use to study how and why these things

happen?

GG: We study the “how”. Our fundamental approach has been genetic. [And the genetic approach] has been an incredibly important approach in general. The *C. elegans* have been the workhorses for this and for a lot of what we know about the mechanisms for development in invertebrates. Or, at least, the molecules that are involved really came from studies in *C. elegans*, *drosophila* and yeast. So, genetics is an approach to understanding a biological problem.

What geneticists do is that they screen for mutants that are defective in the process that they’re interested in. Then, from the mutant phenotypes, they first try to figure out how the mutation effects function and if the mutation recessive or dominant. If it’s recessive, has it partially or completely lost the function of the gene? If it’s dominant, how has it messed up the function of the gene? [Then, what the gene normally does is inferred by how the gene function is affected by the mutation.] If your screen works well, you’ll have a number of genes involved in the process.

You move on from there [by looking at] which molecules were encoded by the genes and what those molecules are doing in terms of cell biology. So, sometimes you hit molecules that you have no idea what they’re doing biochemically and those are the hard ones. But those maybe are the more interesting ones!

So, that’s kind of an initial approach into the problem. There are lots of people who figured out apoptosis in *C. elegans*, and that is what they did. They looked for mutants that were defective in apoptosis. Randy Schekman studied secretion and the initial thing that was done was a screen for temperature sensitive mutants that are defective in the ability of yeast to secrete proteins. So, you can just go through the list of the different processes that people have studied.

BSJ: What about newer techniques based on RNAi (RNA interference) and genome wide studies?

GG: So, we have done RNAi screens but there are qualifications associated with it. Sometimes it doesn’t work (due to off target effects) and, actually, it doesn’t work particularly well in neurons. In *C. elegans*, though, they tend to be quite specific and there are not too many off target effects compared to other organisms. And it’s really easy to do in *C. elegans*. So, yes, it’s a valid approach but sometimes it doesn’t work or you get very, very weak effects. That is, you don’t knock down the function... Some genes are really sensitive to dosage effects so if you just reduce them by 70-80% and generate a phenotype. But there are other genes where you need to get rid of 90% of the function to see a phenotype. So these would be more impervious to RNAi.

But there has been resurgence in screen approaches just using genetics because of the ability to quickly identify mutations using whole genome sequencing. It used to be that it would take a huge amount of time to find the mutations... now it’s much easier!

BSJ: In regards to the various proteins involved in asymmetric cell division, what do we know about the

model right now in terms of where the known players fit in?

GG: We’ve identified a lot of components... There are the Wnt signaling pathways involved in many different aspects including migration and asymmetric division. The Wnt signal seems to polarize the cells and that’s actually understood very well in *C. elegans*. What we found is that the Wnt additionally controls the positioning of the furrow (the cleavage furrow). How the Wnt controls that is not understood but there are these protein kinase pathways and membrane trafficking pathways that we’ve identified. The idea here would be that these are regulating components of the Wnt signaling pathway.

The other thing that we’re interested in came from this understanding that there are two mechanisms controlling the asymmetry in cell divisions: Firstly, the spindle can move and that determines the position of the furrow. In other cells, the spindle eventually moves but the furrow forms first. There are a couple of molecules we’ve identified in our screens that control one of these types of divisions without controlling the other. We’re interested in developing the model for how these mechanisms work. Most of the molecules we’ve identified are involved in both, so there are some shared pathways but we do have some molecules that are specific!

This brings up a very interesting question of why the cell is undergoing apoptosis and why there would be two different mechanisms to generate this asymmetry for apoptosis. But we can continue working on the “how” of the system but the “why” eludes us.

BSJ: And how does these models translate to vertebrate and human biology? Would you expect it to be analogous to some extent?

GG: It’s always hard to say for sure but all the molecules we’ve identified have homologs. These don’t necessarily have to contribute to apoptosis... We know of one case whereby the gene isn’t only controlling the apoptotic pathway but also other divisions that are asymmetric. The idea is for these basic cell biology principles to be conserved [through evolution].

BSJ: And perhaps even more broadly, where do you see this field and your research going in the near future?

GG: I would like to figure out how this polarity is established and how the asymmetric division is executed before I retire!

IMAGE SOURCES

UC Berkeley Department of Developmental and Regenerative Biology
https://en.wikipedia.org/wiki/Caenorhabditis_elegans#/media/File:Adult_Caenorhabditis_elegans.jpg
https://lh3.googleusercontent.com/-so4u_5C_YNEgRq9K5E6fjq2qHOH4HI07CvchHPk7Uc2qnaaJ5A7qwnRNQK3wS3i7PgOx=s170
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Layout by Jacob Ongaro

AN INTERVIEW WITH PROFESSOR KENNETH RAYMOND ON SUPRAMOLECULAR CHEMISTRY: SYMMETRY BASED CLUSTER FORMATION

Manraj Gill, Yana Petri, Tiffany Nguyen, Sabrina Berger



Figure #1. Dr. Kenneth Raymond, Chancellor's Professor of Chemistry

Dr. Kenneth Raymond is a Chancellor's Professor in the Department of Chemistry at University of California, Berkeley. Professor Raymond has been interested in a variety of topics in bioinorganic chemistry and coordination chemistry. In this interview, we focus on one of his specialties, the assembly of highly symmetric supramolecular clusters. We discuss not only the role of symmetry in the formation of such molecular structures but also the application of these clusters in catalytic chemistry.

Berkeley Scientific Journal: How did you get involved in research in chemistry?

Kenneth Raymond: I liked chemistry since I was 12 years old. I was 12 years old when I got my first chemistry set. My mother thought I was too young when I wanted it two years earlier. In those days, real chemicals came in those chemistry sets! In high school, I had a really good chemistry teacher who also taught physics. He let me have free run of the lab for making standard solutions. Aside from almost killing myself a couple of times, that was a really good experience!

Also, it got me into Reed College, which turned my life around. In my first two years of high school, I had a math teacher that was sort of egg shaped and wore these purple dresses. She would be up next to the chalkboard and would get this perfect white ring around her. And she looked just like an Easter egg. She thought I was rude and I'm sure that's true. She gave me bad grades for behavior but all of the people I was tutoring in the class were getting A's. So, by my reckoning at the time, I thought I was winning this battle.

In my junior year, I decided I didn't want to be a juvenile delinquent; I wanted to be an intellectual. And that turned out to be more productive.

BSJ: And was it at Reed that you began focusing on chemistry?

KR: I started doing undergraduate research at Reed after my freshman year. And Reed had this undergraduate thesis. It's up there on the shelf but I won't show it to you, it's too

embarrassing. An undergraduate research thesis was great preparation for the PhD. The PhD was almost easy by comparison. My best friend at Northwestern Graduate School and I were probably the two best-prepared students. He was from Harvard; I was from Reed. So I was in a hurry; I went straight from graduate school to my job here. I have never applied for a job in my life!

BSJ: Really?

KR: It was a different world. My PhD supervisor was a very well known inorganic chemist at Northwestern.

He pulled me into his office at the end of my second year and said, "Well Ken, things are going fast for you this year. What do you want to do in the future? Not industry right?"

I said, "I don't think so."

"Not the national labs?"

"No."

"So you want to be an academic?"

"Yeah, what do I do?"

"Don't worry I'll take care of it."

Next thing you know, I get a phone call from Caltech, Berkeley, and Riverside. So I went off to give talks. Harry Gray, who just turned 80, introduced me at my interview at Caltech and I was so nervous—I had just turned 25. I got up and said, "It is very nice to be here at MIT." True story! He thought it was a joke and everybody laughed. Things got easier after that and I got the job of my dreams and I kept it. Very dull job history; I've been here my whole career!

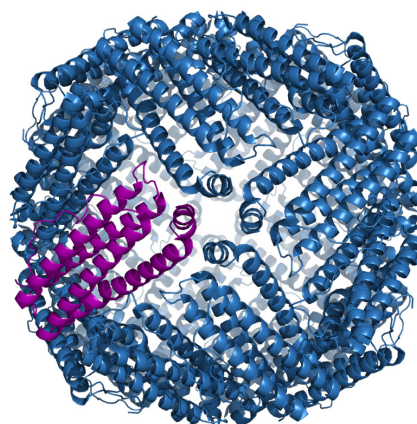


Figure #2. The molecular structure of ferritin

BSJ: So what is supramolecular chemistry and how did you first get interested in it?

KR: For me, it's a relatively recent interest. It only goes back 20 years, probably as long as you have been on the planet. I had a long-standing research interest in biological iron chemistry, especially transport and storage. The way we store iron is in ferritin. Ferritin is a supramolecular protein. It always has exactly 24 subunits, never 23, never 25. It has high octahedral symmetry.

One day I was staring at this in new kind of way. How does this work? I looked at the crystal structure in some detail. It was already an accurate structure and you could see a four-fold interaction site, a four-fold octahedron. There are hydrogen bonds, hydrophobic interactions and so forth. All of which add up to a substantial interaction. But its direction is like a lock and a key where the lock and key are 90 degrees apart. So that forms a tetramer with four-fold symmetry. Elsewhere on the protein, there's a three-fold interaction site. Now, the lock and the key are 60 degrees apart. That says, "Form a trimer with three-fold symmetry." So, how do you do both of these? You make the angle between those interactions equal half the tetrahedral angle: the magic angle of the cube, 54 degrees. The only thing that can form is a 24-mer with octahedral symmetry.

So I had two thoughts at the time... One was, "This is obvious, I must be the last person on the planet to understand this." But if you look in the literature, there was nothing in the description like I just gave you! So, the second thought was less pleasant, "This is nonsense, you're fooling yourself." But, if it's real, it's a recipe for how to make things. So I set about to make clusters where the interactions are not hydrogen bonds, but metal-ligand interactions. Those are directional, rather strong, and are reversible! That's really important, that's a key to supramolecular chemistry.

It's like a Lego set: there are a million ways to put it together in the wrong way, but only one correct solution. So, in the case of supramolecular clusters, if you make a mistake in linking things, you have got to be able to back out of it. That got me started. One of the early clusters we made has been like the Energizer Bunny: it just keeps running! And we keep discovering that it does new things. Our current record for catalyzing a reaction is a 20 million-fold rate enhancement (relative to the uncatalyzed reaction).

BSJ: Were other symmetric biological clusters similar to ferritin known at the time?

KR: Yes, protein viral capsids with icosahedral symmetry. Icosahedral symmetry has 60 symmetry operations.

BSJ: Why was the initial interest only in ferritin?

KR: I wasn't trying to reproduce ferritin. The same analysis and argument would work for the protein viral coats. Why are they 60mers or 120mers or 180mers? There may be three different proteins, they trimerize and then 60 of those trimers get together to form the viral capsid. The simplest capsids are for bacteriophages. The virus that gives you a cold uses a bigger cluster to hide its nucleic acid. But in each case,

it's a question of how to package nucleic acid inside a robust protective coat.

BSJ: Talking more fundamentally, we read this chapter you wrote in the book, *Beauty in Chemistry*...

KR: I hope you enjoyed it. I had fun writing it! You wouldn't know this because you don't know the whole field, but these are some quite prominent supramolecular chemists.

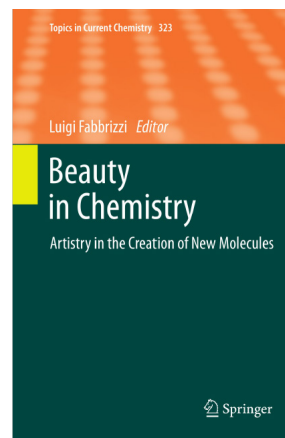


Figure #3. 'Beauty in Chemistry: Artistry in the Creation of New Molecules'

BSJ: Could you elaborate on how chemical synthesis can be regarded as beautiful?

KR: Well, behind you is a supramolecular structure called the quartz crystal. Now, that's only supramolecular in the interior of the Earth at very high temperatures. In other words, you can crystallize it under equilibrium conditions. It's way too cold to be reversible now, but it's chiral of course.

How do you take SiO₂, just a chunk of silica, and make a chiral structure out of it? Well, it crystallizes in spirals and half the time they're right-handed and half the time they're left-handed. Once the crystal starts, if it grows perfectly, it's all the chirality. And it's beautiful, right? Why do we like gem stones? Because of their beautiful colors; but also because they have these faceted surfaces and they scatter light.

BSJ: Do you think what exactly underpins beauty in chemistry would be some degree of symmetry?

KR: I think so. I'm sure there are as many opinions on that as there are chemists, but I think so. Well actually, I think a very large subset would agree with me. That was the whole point of this book. Symmetry is terribly important in all kinds of areas. There's a wonderful book, *Symmetry and the Monster*. It talks about 10⁵² symmetry elements. That's a number that if you started counting, and you could count really fast, it would be the end of the universe before you got to that number.

BSJ: You've already briefly covered this, but how would you explain the mechanisms behind supramolecular clusters to the general audience?

KR: Well, the most interesting thing I think is that it is a way to make complex structures out of simple subunits, and nature uses it that way. I mean these viral capsids are huge, but they're composed of much smaller individual proteins. It's potentially

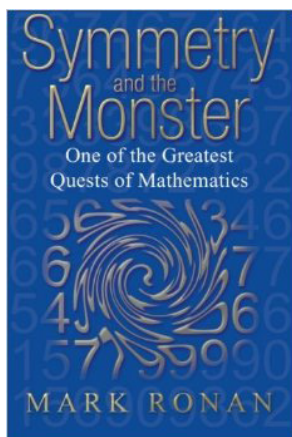


Figure #4. 'Symmetry and the Monster: One of the Greatest Quests of Mathematics' by Mark Ronan

a way to manipulate and do new kinds of chemistry. In my view, chemistry is increasingly becoming more complex, quite in the same way that people think about complex mathematics. Nature is as good an engineer as she is a chemist. A big part of cells is that they are not just single solutions with just one membrane around them. There are all kinds of little subunits. And those structures are terribly important to the functioning of the cell.

My collaboration with Bob Bergman started off 10 or 12 years ago, he's famous for carbon hydrogen bond activation. That's a fundamental chemical bond. Most of that chemistry is very important for things like catalysis in the chemical industry. It involves organometallic compounds, which are metals with carbons around them, and typically non-aqueous solvents such as toluene or whatever. But his catalysts are monocations, and they're greasy. Well, that's the kind of guests that our clusters like! So it occurred to me, let's try putting some of Bergman's catalysts inside our cluster. So we can do non-aqueous chemistry in aqueous solution, and it worked!

That's a kind of green chemistry. Things went on from there, but that's how our collaboration started. I ended up doing a lot of organic chemistry that I certainly would've never dreamed of, and this comes from Bergman and Dean Toste. That's the best kind of collaboration. None of us would have done this individually.

BSJ: To what extent is the spontaneity in supramolecular cluster formation due to symmetry?

KR: What I gave you is ultimately a symmetry analysis. The trick is how to force the molecule to go in the direction you want it to go, not in other directions. We designed our ligands with a two-fold interaction site and a three-fold interaction site because the tetrahedron has symmetry numbers two and three. We designed it so that those axes could only be about 54 degrees apart. It's a rather rigid ligand system; it's very planar. And that's what drives the cluster formation.

BSJ: Would you say that the symmetric state is the lowest energy state?

KR: Not automatically. You have to design it that way.

BSJ: But once the cluster is formed, it would be satisfying the lowest energy?

KR: Yes, exactly. That makes it the lowest state. Each metal wants to have three of these catechols around it so for the specific cluster, and each of the ligands wants to have both of its ends coordinated to a metal so there are no loose sticky ends. That then makes it the lowest energy state. But I interpreted your question as, "In a very general way, is the most symmetric structure always the lowest energy?" I would say no. You have to build it that way.

BSJ: Perhaps even more fundamentally, what do we know about accounting for symmetry in thermodynamics. Can it be quantified?

KR: Well, in physics it's terribly important. All these string theories are dealing with multidimensional spaces and symmetry between particles and antiparticles and so forth. It's very important. It's also terribly important in chemistry, in that, for chemical bonding there are wave functions. Those wave functions of an atom have required symmetry. Any wave function, whether it's a guitar string or a hydrogen atom, the different wave functions are orthogonal to each other. That's why if you pluck a guitar string, you may hear a transient note for a minute, but then the note that continues is a single tone. You can make a harmonic of it, that harmonic is orthogonal to the fundamental. Same thing happens with the atomic wave function. Symmetry is very important there because it helps you analyze the quantum mechanics so all of the theory behind bonding.

BSJ: But once you have the ΔG° free energy, can you account for symmetry in that regard? That is, the reaction being driven purely due to a symmetric reason?

KR: I think not as easily in thermodynamics as in quantum mechanics. In fact, I started a course here years ago on chemical applications of group theory. Group theory is a mathematical application of symmetry.

BSJ: What does that tell us about the evolutionary selectivity for symmetrical structures?

KR: That's a great question, and people are still arguing about that. [At the most fundamental of levels], the neutrino is chiral. When it travels through space it can have a spin this way or this way.

BSJ: In regards to having a host system, why do the clusters have to be symmetric? Does it relate to the need for repeated assembly or dissociation of subunits?

KR: Well, let's suppose that, instead of one identical ligand, all six were different. How many different isomers will there be, how many products will there be? It will be an awful mess! In order to have one simple thing, you have to have symmetry and make all of those ligands equivalent. Nature does the same thing.

BSJ: But if you had to catalyze a different reaction that required different space in the host... Why would it be beneficial to rely only on symmetry in terms of formation of symmetric supramolecular structure compared to making some other non-symmetric cluster?

KR: Why, yes! But even if I'm making another cluster, I want it to be one thing, not a thousand different things. So if I make another cluster, I want all components to be the same, because otherwise I do not know what I have in solution.

BSJ: We were wondering what exactly happens at molecular level in the cluster. Is the cluster being formed around the host, or is the host being taken in?

KR: Usually, the latter. We do what we call "templating" in some cases, where the guest interaction will help to form the cluster. Usually, we make the cluster separately and then have the guest as a reactant. Most of what we've been doing recently is using the cluster like a little enzyme. It is chiral, but the only thing chiral about it is the structure. We can have a guest come in, catalyze their chemistry, and then it's important that the products are not good guests. Because you want to spit the products out and go around the cycle again. We've shown in several cases, that it is really an enzyme-like mechanism that follows Michaelis–Menten kinetics.

BSJ: So, it is not like the cluster is completely encapsulating, but more that the substrate is sitting on an active site in an enzyme?

KR: It sort of is. Remember, the trick is to get the angle between the three-fold axis and the two-fold axis fifty-four degrees? We initially did that by a lot of computer design and testing different structures. The three-fold axis is built in, because these are metal ions, so these are six-coordinate. The two-fold axis we built into the ligand. It has structural memory. The chirality of this vertex is random, initially, but ones we set it, all the other vertices, because of planarity, have to have the same chirality.

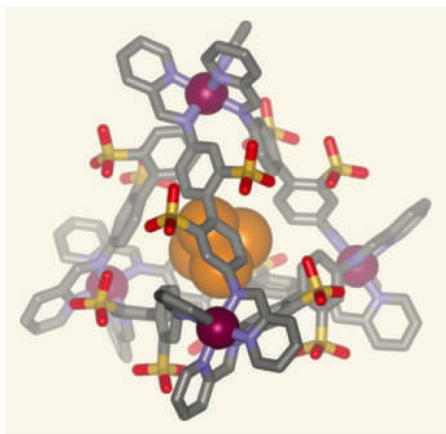


Figure #5. Structure of the host-guest complex

The cluster is very soluble in water because it has a twelve minus charge; each vertex is a three minus charge. It is chiral, because of this twist around the metal center. We can resolve them, and more recently we've been making chiral versions that have chiral substituents. There is an inside and an outside to the molecule, because the six naphthalene rings coat the inside. The inside is completely hydrophobic.

So the question is, how is the young Schwarzenegger like supramolecular chemistry? This was kind of the rap for

the last twenty years. And the answer is: they both look good but do not do anything! So, we have been trying to disprove that. We are trying to do things inside the cluster.

BSJ: Does it matter, in your case, if it forms an enantiomer?

KR: We can get enantiomeric excess from reactions that start with a non-chiral substrate. The only thing that is chiral, [in that case], is the flask in which we are doing the reaction!

Manraj: How exactly are you directing it towards one conformation?

KR: To be honest, I can observe it, but I can't predict it. I wish I could. You ask about how the exchange occurs and the dilation of the aperture by twisting the naphthalene rings is a pretty low energy process. Initially, I thought that we must be breaking a bond up here but that takes a fair amount of energy. So, this process is fast on the laboratory scale, milliseconds, but slow on the NMR time scale. That's how guest exchange occurs. If you have two molecules of the same volume, one that looks like an American football, and the other that looks like a soccer ball, the American football is faster going in and out, because the dilation of the aperture is smaller.

BSJ: So it clearly is very flexible?

KR: It is, and of course that is true with enzymes, too. People have often made mistakes. You know you look at a potassium channel protein and say, "Oh, potassium could never fit through there." But that's nonsense, because they are rocking and rolling all the time, the structures are very dynamic.

BSJ: Why does the tetrahedral confirmation specifically often underlie supramolecular cluster formation?

KR: The simplest of the polyhedra is the tetrahedron. So, I thought I would start simple. Now, one thing we tried to make early on was an octahedral symmetry. And you haven't seen that because it didn't work! That doesn't mean it never will work but that the approach I was trying didn't work. We're interested in doing that, though. I have a couple students who are working on expanded clusters and different cluster designs. For example, a tetrahedral cluster where the ligand occupies the face of the tetrahedron.

So, it has three bidentate [directing] groups and so the stoichiometry would be four metals and four ligands, instead of 4:6. And it's easier to extend that. It's easier to make it bigger. Of course, the longer you make the ligand, the volume goes up as the cube of the extension of the length. So, we can make bigger clusters.

BSJ: So, is the tetrahedral the largest working cluster you have created so far?

KR: Yes

BSJ: Is that, then, a limiting factor in the type of reactions you can catalyze?

KR: Well, there are lots of people who are making supramolecular systems. Our system is unique in that it is inherently chiral. And that means that you can catalyze chiral reactions. And in the last year we discovered that we can do photochemistry and electrochemistry inside the cluster. So, this thing is going off in new directions. Until all the gold is

mined out, I'm going to keep mining it!

BSJ: What are some of the ways in which the reaction rates [of the catalyzed reactions] can be accelerated?

KR: Well, I'll give you an answer that is not really an answer. We must be binding the transition state. Do I know what that looks like? No, I don't really.

I'll give you another example. What you learn in beginning organic chemistry is that SN1 reactions racemize and SN2 reactions, that have a chiral carbon center, invert the absolute stereochemistry. But, we have an SN2 reaction that retains the absolute stereochemistry.

How does that work? Well, we have a chiral molecule. If you do hydrolysis in water, you get 84% retention of the chirality. If you catalyze it inside the cluster, you get 74% retention. And how does that work? I can give you two limiting explanations... If the leaving group goes off, we get an SN1 reaction. But it's snuggled up next to this naphthalene, a pi complex that is not free to rotate. So, now when the new entering group comes in, it comes in from the same side and we get retention of stereochemistry.

Now, I'll give you the opposite extreme: SN2. The naphthalene acts as a nucleophile and displaces the leaving group. Now entering group that comes in, displaces again. So, in fact we've have two SN2 reactions! Which is true? Well, how do they differ? They differ in the transition state! In SN1 this would be something like a 3-3.5 angstrom (Å) distance. If it's SN2, it'll be more like 1.5Å. Pretty big difference! But I can't see the transition state. So, the only way we're going to be able to answer this is theory.

BSJ: We talked about some of the promising developments already but where do you see the research in supramolecular chemistry going?

KR: Lots of things, I think! Almost all of my career, I've shamelessly stolen from nature. She has no patents; she has no copyrights! So, I look to nature. What does nature use supramolecular systems for? To deliver things; to protect things. So, drug delivery systems may be an application. To catalyze things. So that would be my speculation for the future.

BSJ: And in regards to green chemistry and environmental chemistry, how would the supramolecular systems concept be used to remove harmful species? Would the structure be able to capture such species?

KR: At the moment, it better be a really expensive toxic species. These are not cheap molecules. So, you cannot be talking about carbon sequestration. That's too high volume and too cheap. But back to the delivery idea, it's very hard to get drugs across the blood-brain barrier, and yet, our antibodies get across that barrier all the time. And they're really big! How does that work and can you mimic that process? Chad Mirkin at Northwestern has shown that you can coat gold particles with DNA and they go into cells. The gold molecule is huge, but it goes into cells. So, all kinds of new methods of delivery and transport might be enabled by this.

BSJ: If you were to use it as a drug delivery system, how would you control when the host is released? But can you

direct the intake and outtake of the host?

KR: I don't know, that's your job! I'm just spinning ideas right now. Those are all problems that will need to be solved but I think we're already at the stage where these kind of ideas have real applications. Does that mean they're going to be easy to do and solve all problems? No.

BSJ: Where does the equilibrium lie in these reactions?

KR: In the reactions we are catalyzing? We're going downhill in energy but we're getting it up over the transition state a lot faster than it would have while just sitting in solution. But in our photochemistry, we're making an higher energy molecule. So, we have shown that the cluster absorbs light and then it can deliver that light energy to a guest inside it, if it is an appropriate guest. That triggers a rearrangement and forms a higher energy product. It's a solar energy conversion...

BSJ: Like a photosystem...

KR: Yeah! And we're arguing that the DOE should give us more money for this. Most people when they talk about photochemical energy are talking about something that you're going to burn as a fuel. Those are really cheap. So, our chemistry would really never be useful for that because it is too expensive. But if you're making high value chemicals, then it makes sense! It's catalytic and using the sunlight.

BSJ: Why is it so expensive?

KR: They're kind of hard to make; ask my students! My hero in science was, and is, Linus Pauling. I met him a few times... two of his children went to Reed and I have been at cocktail parties at his house! He was so brilliant on so many different things. He's the one who said that enzymes bind what today we would call the transition state. And that's what we're trying to do. I want to be able to do some theory on these so that I can predict the next reaction we can catalyze rather than just try one and see if it works.

BSJ: Thank you very much for the time, professor!

KR: Nice to meet you all and good luck with your Berkeley careers and thereafter!

IMAGE SOURCES

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EXTRACTING INFORMATION: CHARACTERIZING NEURONAL CELL TYPES IN THE GP_H BY THEIR ACTIVITY PROFILE.

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ABSTRACT

The globus pallidus is a major output station for the basal ganglia, a subcortical region of the brain that is heavily implicated in action selection and decision making. A subpopulation of neurons in the internal segment (GPi) projects to the lateral habenula (LHb), often associated with the limbic system and known to encode for negative motivational value. Dysfunction in these structures have been implicated in neurological diseases, such as depression and schizophrenia, which are ultimately disruptions in the ability to evaluate environmental cues and regulate motor output. In order to gain more information about the neurons which encode for this behavior, we conducted extracellular recordings while the mice are carrying out a set of reward learning tasks and analyzed the collected spike trains. We detail here the methods of information extraction from the neuronal populations that we have classified. We also present preliminary results of their activity profile for various outcomes as well as for reward history and prediction error. With collection of information from a larger set of cells, we might be able to more definitively gain an understanding of the methods by which these neurons encode motivation, action selection and outcome evaluation.

INTRODUCTION

How is an organism motivated to act on a value-based decision? How does an organism learn to perform a specific task, to make an appropriate choice, to make the right associations? How do neurons encode initiation of movement and assign values to an expected outcome?

Basal Ganglia Circuitry

Circuits governing reward and decision-making have been the subject of extensive studies.¹ The basal ganglia emerges as a structure of primary importance in action selection and outcome evaluation, in addition to its role in motor control via the more established basal ganglia-thalamocortical circuitry.² The thalamus forms the central core of the brain and may be divided based on the spatial location of various nuclei, which receives input from distinct pathways and projects to well-defined cortical areas. Many critical functions such as sensory and motor mechanisms and cognitive functions are relayed via the thalamus. Specifically, the medial nucleus sends projections to the prefrontal cortex and is heavily associated with higher cognitive functions, whereas the ventrolateral nucleus of the thalamus sends information to the motor and somatosensory cortices and is associated with motor tasks.³

The primary function of the basal ganglia is likely to control and regulate activities of the motor and premotor cortical areas so that voluntary movements can be performed smoothly.^{4, 5} Stimulating the motor cortex of monkeys at various locations results in stereotyped sequences of movements. Thus, motor control may require the activation of these elemental motor programs in the precise temporal

order to accomplish a sophisticated motor plan.^{6, 7} These motor programs involve inhibitory networks across various cortical and subcortical structures such that a release of this inhibition permits a motor system to become active.⁸

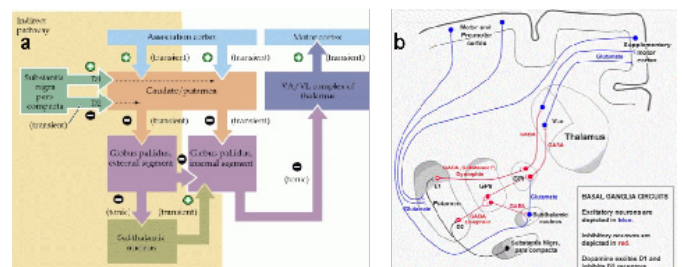


Figure 1: Basal ganglia circuits. a. Block diagram of circuits.⁹ b. Schematic of circuits connecting various basal ganglia nuclei.¹⁰

The basal ganglia is comprised of the striatum, consisting of the caudate and putamen, the internal and external segments of the globus pallidus (GPi and GPe), the subthalamic nucleus (STN), and the substantia nigra pars reticulata and pars compacta (SNr and SNc).³ The striatum is the input center of the basal ganglia, and receives excitatory afferents from the cerebral cortex such that along the extent of the caudate and the putamen, inputs from cortical regions vary by their relative proximities. In particular, the primary motor cortex projects mainly to the putamen, and the topography of projections is maintained in the intrinsic circuitry of the basal ganglia. The GPi serves as the major output station of the basal ganglia, along with SNr. These structures are tonically active, and impose inhibitory afferents onto the thalamus, which relays excitatory signals onto the

primary cortex associated with the relevant thalamic nuclei.⁶

There are two distinct pathways in the basal ganglia: the direct pathway, which has a net excitatory effect on its targets and the indirect pathway, which has a net inhibitory effect on its targets (Fig. 1) A model for achieving an appropriate motor response to a task is that these direct and indirect pathways coordinate the execution of stored elemental motor programs, and an upset of this balance results in motor dysfunction.

The direct pathway begins with medium spiny neurons of the striatum, which send inhibitory projections to the GPi and SNr and serves to release upper motor neurons from tonic inhibition, thus activating them. The indirect pathway starts with another population of medium spiny neurons sending inhibitory inputs to the GPe, which lifts the tonic inhibition on the excitatory neurons of the STN projecting to the GPi. The GPi is activated and the level of tonic inhibition is increased.⁹ The direct and indirect pathways create a complex sequence of excitation, inhibition and disinhibition. The direct pathway is a positive feedback loop that has a net excitation of the motor cortex, whereas the indirect pathway is a negative inhibitory feedback loop, and a delicate balance is necessary for adequate performance of various motor tasks.³

In addition to the direct and indirect pathways, the nigrostriatal pathway, connecting the SNc to the striatum, has complex inhibitory and excitatory effects on striatal neurons. This pathway is composed of dopaminergic neurons in the SNc and largely GABAergic neurons with dopamine receptors in the striatum: activation of D1-like receptors on striatal neurons in the direct pathway induce adenylyl cyclase-dependent increase in intracellular Ca²⁺, which stimulates neurotransmitter (GABA) release by the medium spiny neurons in response to dopamine, whereas indirect pathway striatal neurons possess primarily dopaminergic D2-like receptors, which inhibit AC activity, and thus inactivates the neuron (no GABA release).¹¹ Thus, activation of the dopamine neurons of the nigrostriatal pathway activates the excitatory direct pathway but inhibits the net inhibitory effect of the indirect pathway.

Encoding Action Selection by Reward Learning

Previously we described the basic basal ganglia circuitry for the activation of a specific motor program and inhibition of competing motor programs (primitive programs stored in the cortex) for a precise sequence of movements that allow the organism to adequately respond to certain environmental cues. However, one can imagine that in order to perform at maximum efficiency, a system has to adopt some measure of learning, so that a familiar cue immediately calls up a stored motor plan, which when executed yields an expected set of rewards. In the case that the outcome deviates from expectation, the system should also have methods for evaluating this error in reward prediction and perhaps adjust its established motor plan. When faced with choices, the

system should also learn to assign values to these various choices based on expected outcome. Thus, behaviors should be affected by rewards, undergoing long-term changes when rewards are different than predicted but remaining unchanged when rewards occur exactly as predicted. Dopamine neurons in the substantia nigra are believed to be involved in reward-dependent behaviors, especially with the reinforcement mechanism involved in learning.¹² **They were found to encode reward prediction errors and were activated by the receipt of an unexpected reward and inhibited by an omission of an expected reward.** Dopamine neurons were also activated by rewards during early trials, when errors were frequent and rewards unpredictable, but activation was progressively reduced as performance was consolidated and rewards became more predictable.¹³

Returning to the nigrostriatal pathway, when activated the pathway has a net excitatory effect on the cortex. As previously described dopamine neurons were activated during the early reward learning stages as well as upon deviations from expected outcome. As such, the changes in pattern of dopamine firing does not simply increase or decrease movement, but rather fine tune the balance of the direct and indirect pathways, allowing for enhanced activation of the cortical motor programs responsible for producing rewarding outcomes and suppression of motor programs that do not result in reward.

The current model of motor movement is that many primitive motor programs are stored in the cortex, and the role of the basal ganglia is to release and inhibit these primitive motor actions in a precise temporal sequence³ such that competing motor programs are suppressed by the indirect pathway and the appropriate program is disinhibited by the direct pathway.⁶ Glutamatergic release by cortical neurons onto MSNs in the striatum with D1 receptors result in a GABAergic output to the GPi, thus lifting its inhibition on the thalamus and eliciting an action. Glutamate release onto MSNs with D2 receptors induces GABA release onto the GPe, thus allowing the STN to excite the GPi and SNr, inhibiting the thalamus and inhibiting action. However, dopamine release by the SNc onto MSNs in the putamen is rather more subtle. It appears that dopamine does not directly induce or inhibit firing in a cell. Rather, its release modulates the excitability of the neuron to glutamate.¹⁴ It has been shown that dopamine D1 receptor signaling enhances dendritic excitability and glutamatergic signaling in striatonigral MSNs, whereas D2 receptor reduces the excitability of postsynaptic neurons to glutamate and release of glutamate by the presynaptic axon terminal. When D2 receptors are activated by dopamine binding, the excitability of the neuron to glutamate is greatly reduced, thus it does not release GABA onto GPe, and disinhibits a competing pathway that was previously inhibited by the indirect pathway. When D1 receptors bind dopamine, the neuron's excitability is enhanced and the direct pathway is more likely to be activated. This piece of information corroborates with the previous observations that dopamine

neurons in the substantia nigra were more active during the learning period of earlier trials, to respond to prediction errors and unexpected rewards. During early trials when an organism is starting to associate a certain motor program with a certain cue in order to receive a reward, release of dopamine onto D1 neurons in the striatum increases their excitability to signals from the cortex and thus promotes the activation and thus consolidation of an appropriate motor program. When an organism makes a prediction error, excitability of D2 neurons is reduced and thus competing motor programs are disinhibited and the mouse can explore different motor programs. The end result of such a modulatory system is that motor habits which are likely to result in reward are retained, whereas those which interfere or reduce the likelihood of reward are inhibited.³

By modulating the elementary motor programs, mid-brain dopamine neurons are key components of the brain's reward system. But how do the dopamine neurons know when to release dopamine to modulate the direct and indirect pathways? It had been unclear which brain areas provide dopamine neurons with the signals necessary for their response to sensory stimuli predicting reward until recent efforts identified the lateral habenula (LHb) as a major candidate for a source of negative reward-related signals in dopamine neurons of the SNc and ventral tegmental area (VTA).^{15, 16, 17} Habenula neurons were activated by a no-reward-predicting, or a punishment-predicting target and inhibited by a reward-predicting target, especially when they were less predictable, whereas dopamine neurons were excited and inhibited by reward-predicting and no-reward-predicting targets, respectively. These results suggest that **LHb sends inhibitory input to dopaminergic neurons** in determining their reward-related activity, and has the potential to adaptively control both reward-seeking and punishment avoidance behaviors.^{2, 18, 19} The positive reward prediction error encoding by dopaminergic neurons— activation and release of dopamine upon an unexpected reward and inhibition of dopaminergic neurons upon an unexpected lack of reward— allows for actions that result in reward to be reinforced while inhibiting actions that no longer result in reward. Increases in habenula activity correlated with the Hamilton Rating Scale for Depression.²⁰ This may be due to elevated activity of the LHb inhibiting the midbrain dopaminergic systems, resulting in decreased drive to seek rewards, which may contribute to depressed behavior.

What else is missing from the perspective constructed of the motor loop of the basal ganglia? The direct pathway releases the tonic inhibition imposed by the GPi and SNr on the thalamus, and activates a certain motor program and the indirect pathway increases the inhibitory activity of the GPi via the GPe and STN, which inhibits competing motor programs. Both pathways are modulated by dopaminergic neurons in the SNc, which is subject to inhibition by the LHb. How does one close this loop? How does the LHb receive

feedback to ultimately induce dopamine release to reinforce a motor program, or inhibit dopamine release in response to an omission of expected reward?

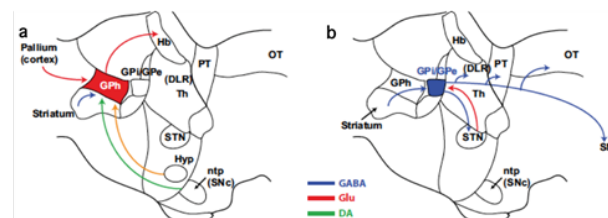


Figure 2: Globus pallidus circuits.²¹ a. GPe receives GABAergic input from the striatum and projects glutamatergic output to the lateral habenula. b. Primarily GPe and GPi neurons participate in the direct and indirect pathways of the basal ganglia, receiving glutamatergic input from the STN and projecting GABAergic input to the thalamus.

The component of this circuit that provides the signal to the LHb appears to originate in the pallidal region, close to the GPi. The GPi is classically considered to be related to the sensori-motor basal ganglia.²² It receives inputs from the dorsal striatum, GPe, STN, and projects to the LHb and the ventral lateral thalamic motor nuclei, which, in turn, innervates the premotor and supplemental motor cortex and, thus, completes a basal ganglia-thalamocortical loop. Pallidal neurons projecting to the thalamus and those projecting to the LHb constitute separate neuron populations.²³ Some fibers originating in the GPi arborize extensively in the LHb, and exhibit numerous terminal-like specializations consistent with an important pallidal influence on lateral habenular functions. It was shown in lamprey that a separate evaluation circuit regulates habenula-projecting globus pallidus (GPh) neurons. These neurons are located in close proximity to components of the circuit that participate in the direct or indirect pathway and can thus integrate real time signals from the cortex to convey to the lateral habenula to call upon modulatory signals. They receive inhibitory input from the striatum but have glutamatergic output and can drive the activity of the lateral habenula, which then inhibits midbrain dopamine neurons in the SNc. The release of dopamine can provide feedback that reinforces a particular motor program.

What are the physical implications of such a network? From what we already know about the dopamine circuit and the communication between the GPh and the lateral habenula, we can derive the expected neural behavior of GPh neurons with various stimuli. When the GPh is inhibited, LHb does not extend inhibitory afferents to SNc and dopamine release onto the D1 and D2 receptors in the neurons of the striatum serves to lift the inhibition by the GPi and SNr and generate a net reinforcement of the motor plan. This should occur in tandem with the direct pathway so that when the cortex sends glutamatergic input to D1 neurons in the striatum, disinhibiting the GPi and executing a selected motor plan, the pathway is reinforced by activation of the dopamine pathways. On the other hand, activation of the GPh allows LHb to inhibit the dopaminergic neurons in the SNc, which

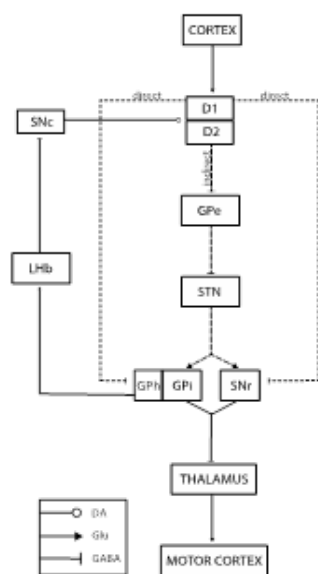


Figure 3: A rough sketch of the components of the basal ganglia: medium spiny neurons expressing either dopamine D1 or D2 receptors make up the majority of the striatum and govern the activation of the direct

occurs upon an omission of an expected reward. This should occur alongside the indirect pathway, so that activation of D2 neurons inhibits the GPe, allowing tonic activity of the STN to activate the GPi and SNr, inhibiting thalamic/motor output. When the indirect pathway is activated, inappropriate motor programs are inhibited, or diminished, and signals from the GPh are conducted to inhibit the firing of dopamine neurons. Note that while both D1 and D2 neurons are activated by the cortex, it is the presence of dopamine signals that determine whether they release GABAergic output. We can then isolate the components of the circuit that are active at the activation of the various pathways.

Thus we have more comprehensively outlined the circuit that coordinates movement such that information about the external world is received in the prefrontal cortex, which curates a series of appropriate motor programs, conveyed to the appropriate effector systems via the basal ganglia circuits. Appropriate programs are reinforced and inappropriate ones are inhibited. This information gives us some insight into how an organism learns and how motivation to perform a certain set of tasks is computed: action sets are built largely by trial and error: ones that yield a reward (which can be either physical or psychological) are selected for by reinforcement and ones that fail to yield a reward are diminished by lack of reinforcement. The lack of motivation could be attributed to a deficiency in the reinforcement learning pathway, perhaps an overactive GPi/GPh, such that actions are not executed and dopamine release is inhibited so that even rewarding motor plans are not properly reinforced.

The circuit looks simplistic: there exists other

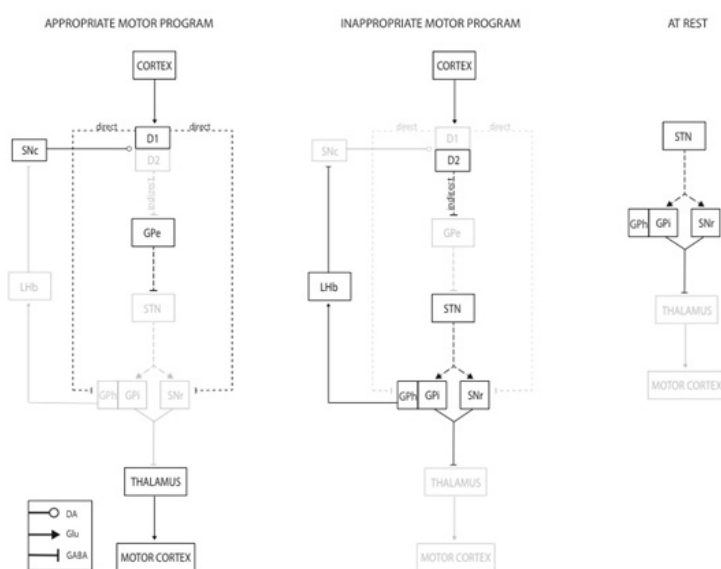


Figure 4: A rough sketch of the current model of activation of appropriate motor programs and inhibition of competing motor programs via the in/direct pathways. At rest, extraneous motor movement is inhibited by tonic inhibition activity of the GPi and SNr.

contributing components to the circuit which have not been incorporated, and whether or not they serve a direct or modulatory function remains to be seen. It is assumed, for instance, that most components of the circuit outlined above have some baseline activity, so that they are active when not inhibited. This may not be correct: there may exist other components that may be actively driving some parts of the circuit (and what drives them?) that are modulated by other parts of the circuit so that there is constant feedback between all components. It was not clear whether the separate population of the GPi that projects to the lateral habenula also receives a distinct set of signals from the STN or the cortex. This is important because it was assumed that the signal that originates in the PFC that sends inappropriate motor programs to the inhibitory pathway is conveyed via the same STN neurons that innervate GPi to innervate GPh neurons, which can then suppress the SNc, in order to coordinate inhibition with dopamine modulation. If this is not true, then the signal to activate GPh neurons must then come from some element in the circuit, perhaps even as far back as the original PFC cortical neurons. How does this other pathway complement the indirect pathway so that dopamine release is timed with inhibition of motion? How do the GPh neurons manage to receive distinct signals from those received by the GPi even though the two populations are so tightly interspersed? More research is needed to answer these questions. The current model of action selection is that signals from the PFC activates a motor program via the direct pathway and inactivates a competing program via the indirect pathway, but how are the direct and indirect pathways distinguished? The medium spiny neurons have D1 and D2 receptors, allowing them to have distinct activation patterns with dopamine release, but how do they get selectively activated to convey glutamatergic

signals form the cortex through either the D1 or D2 neurons? An appropriate motor movement in one context may be a competing motor movement in another; what are the changes that occur for them to be appropriately wired to either a direct or indirect pathway? There are recent reports that both pathways are concurrently activated during movement, and that all MSNs may facilitate or inhibit movement depending on synaptic plasticity.^{24,25} This may not undermine our current model of movement, as a motor program may call upon activating some motions and inhibiting others, but it may add another layer of complexity that is also important to consider.

It is perhaps good to remember some principles of pathways, that the complementation of pathways is important so that competing pathways are not on at the same time, and resources are not spent to activate competing pathways. This implicates the placement of various components in the circuit. Similarly, the brain exists to process external information, which is received via the cortex. Any signal must have originated in and must be ultimately conveyed back to the circuit, if the pathway exists to effect a systemic movement. This raises the larger question as to how information from other pathways is incorporated into the motion circuit: how is sensory information as to the receipt of a reward or punishment conveyed? How does memory about previous decisions factor into the action selection of a motor program? Upon encountering a reward prediction error, in addition to the more immediate update in motor motion, how does the circuit update learning and memory circuits for subsequent decisions? With more information about a circuit comes more questions. We may, in fact, be able to answer the last one, the cross-talk between circuits. We see that the lateral habenula may emerge as an important component in other analogous circuits in the mid-brain, and even a peripheral understanding

of these other circuits can guide one's investigation of the motivation and reward circuit by studying the extensive networks of the LHB, and its afferent pathways.

The lateral habenula also features prominently in another important cortico-basal ganglia circuit: the limbic loop. The key structures in this network are the anterior cingulate cortex (ACC), the orbital prefrontal cortex (OFC), the nucleus accumbens in the ventral striatum (VS), the ventral pallidum (VP) and the midbrain dopamine neurons in the ventral tegmental area (VTA). As in the motor circuit, the thalamus (more specifically the medial dorsal thalamus, MD), is the final relay center that conveys signal from the loop to cortical regions.²⁸ Connectivity between these areas forms a complex neural network that mediates different aspects of reward processing.

Starting with the LHB, inhibitory projections are extended to dopaminergic neurons that reside in the ventral tegmental area (VTA), as depicted in Fig. 5b.¹⁶ The VTA has been extensively studied in reward learning and fear circuits, and shows increased activation in response to stimuli that predict reward. Keep in mind that previously we had also identified the SNc to contain reward-positive dopamine neurons. The VTA, as well, is the seat of dopaminergic neurons in the reward circuit and projects to the ventral striatum, which contains D1 and D2 receptors.²⁹ The nucleus accumbens (NAcc), located in the VS, features prominently in the reward circuit. The VS receives a large glutamatergic input from the OFC and ACC and projects inhibitory input to the VP and to the VTA (Fig. 5c). The ventral pallidum projects inhibitory output to the medial dorsal thalamus, which has projections to the cortex. This is analogous to structures and functions in the motor loop, where the striatum (putamen) at the head of the direct and indirect pathways, receiving

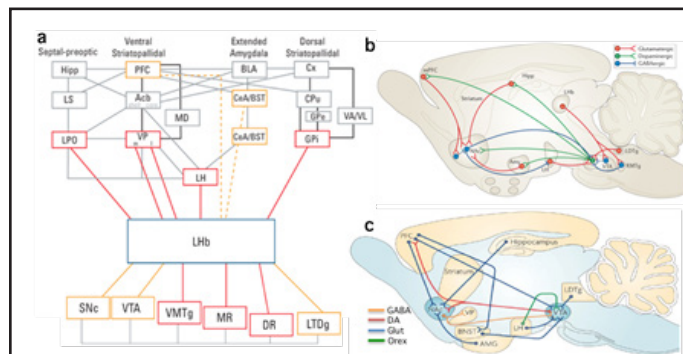


Figure 5: Lateral habenula in reward circuits. a. LHB projects to the SNc and receives input from the GPi, as previously described in the motor loop. It also projects to the VTA, and receives input from the VP, which are components of the limbic loop, suggesting that cross-talk between the two circuits could find a crucial link in the LHB. (The colors represent strength of connection.)¹⁷ b. Activation of the LHB results in GABAergic output to the dopaminergic neurons in the VTA, which sends DA to the NAcc (ventral striatum). The mPFC sends glutamatergic output to the GABAergic neurons in the NAcc.²⁶ c. Note that the NAcc projects GABAergic output to both the VP and the VTA.²⁷

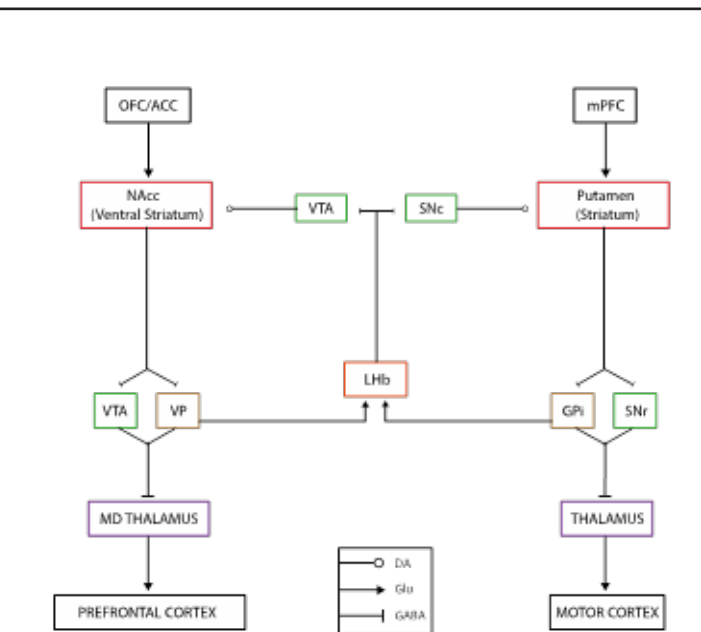


Figure 6: Analogous pathways in basal ganglia circuits. Note analogous functions and physical proximity of structures in same color boxes.

glutamatergic input from the sensorimotor cortices in the motor loop and projecting GABAergic output to the globus pallidus and substantia nigra, which in turn inhibit the ventral thalamus that projects out to the motor cortex.

The proximity of the complementary structures (cortical areas, NAcc/Putamen, VTA/SN, VP/GPi, dorsal/ventral thalamus) probably does not come as a surprise. Reward anticipation induces—and non-reward outcomes suppress—VS activation,³⁰ leading to proposed theories that VS activity tracks a reward prediction error.³¹ The VP also has connections to the SNr and STN, and constitutes a major afferent of projections to LHB (Fig. 5a), in addition to the GPi.^{32, 33} With the LHB having connections and receiving projections to major structures in both the motor and limbic loop, it, along with midbrain dopamine neurons in the VTA and SN assume important roles in the feedback between the two circuits.

The idea that VS (limbic loop) can influence the dorsal striatum (motor loop) through the midbrain dopamine cells originated in rodent studies, which demonstrated projections from the NAcc to the dorsal striatum, through the SN. Through this pathway, therefore, limbic regions could impact on the motor regions of the basal ganglia. The dopamine neurons in the VTA and medial SN are associated with limbic regions, and those in the central and ventrolateral SN are associated with the associative and motor striatal regions, respectively. Taken together, the interface between different striatal regions through the midbrain DA cells is organized in a loop interconnecting different functional regions of the striatum and creating a feed forward organization from reward-related regions of the striatum to cognitive and motor areas.²⁸

Cortico-striatal Loops

It has not escaped attention that in addition to the motor loop, the basal ganglia supports other cortico-striatal loops that are organized in parallel both functionally and anatomically.³⁴ This should come as a relief as we gather information about the learning and motivation circuitry, as to develop an appropriate behavioral response to external environmental stimuli, information about motivation and reward needs to be combined with a strategy and an action plan for obtaining goals. The reward circuit comprises several cortical and subcortical regions forming a complex network that mediates different aspects of reward-based learning, leading to adaptive behaviors.

Simultaneous activation of seemingly unconnected regions (eg. mPFC and OFC) indicate that there must be some communication between the regions. This would likely route through the basal ganglia, which has evolved from a historically purely motor or sensory-motor function to a more complex set of functions that mediate the full range of goal-directed behaviors, including emotions, motivation, and cognition. The idea of separate cortical loops in the basal ganglia was

expanded to include several parallel and segregated circuits based on the finding that each general functional area of cortex (limbic, associative, and sensorimotor) is represented in specific regions in each basal ganglia structure.³⁴

Reward pathways interface with circuits that mediate cognitive function to affect motor planning. Within each of the cortico-basal ganglia structures, there are convergence zones that can link the reward pathway with those associated with cognitive function. Through these interactive networks, information about reward can be channeled through cognitive circuits to influence motor control circuits. Fig. 7b is especially informative in depicting the cortico-striatal-basal ganglia circuits in parallel, with the major players of the circuits shown in their respective locations in the brain. The physical proximity of the structures that serve analogous functions is all the more elucidating. The length and direction of the path of a signal through the structures traces out a well-worn loop, and may have origins in the development of the brain, as structures differentiate and separate. Looking at how these tracts compare in more primitive organisms could be interesting. Temporal coordination could be key: while reward anticipation activates the NAcc in the ventral striatum, reward outcomes subsequently recruit the caudate and putamen, including the supplementary motor area, and most likely involves dopamine pathways. Thus, the ventral cortico-basal ganglia network, while at the heart of reward processing, does not work in isolation: there are pathways that allow communication between different parts of the reward circuit and between the reward circuit and the associative circuits.

Characterizing neuron types in the GPh by activity profile

Armed with the large amount of background information as presented above, it may be wise to have a starting point. The globus pallidus, as a major output center for the motor basoganglia circuits arises as a region that can potentially elucidate the inherent networks. We focus on the neurons within the globus pallidus that have projections to the lateral habenula. The lateral habenula has been shown to have extensive projections to structures in both the motor loop and the limbic system, which raises the possibility that the projection from the GPi to the LHB might be a key link between the basal ganglia and the limbic system, providing reward-related information and initiating motivation to

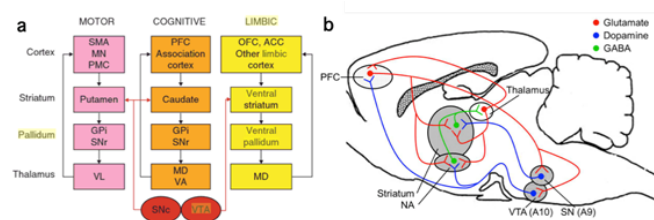


Figure 7: Analogous cortico-striatal circuits. a. Block diagram of circuits.³⁵ b. Schematic of circuits showing the various basal ganglia circuits in parallel.³⁶

move.³⁷ It has been shown in primates that this subpopulation of neurons, referred henceforth as GPh, exhibit negative reward behavior, similar to that of the lateral habenula. The negative reward signal may contribute to the well-known reward coding of neurons, during which constant inhibitory signals from structures in the basal ganglia inhibit various motor neurons. Inhibition of these neurons allow units downward from the circuit to fire and from there initiate various motions.³⁸ To better understand this circuitry, the wiring as shown in Figure 4 must be associated with real-time firing patterns in the neuron population of interest alongside the biochemical profile. We can collect such data by simultaneously obtaining electrophysiological and biochemical information from the same neurons. In addition, as a major objective of neuroscience is to understand how neural circuits give rise to behavior, it is optimal to obtain functional data on neural firing patterns from animals that are consciously perceiving stimuli and can respond consciously.

MATERIALS AND METHODS

Behavioral Training

We classically conditioned mice with different auditory cues that predicted appetitive or aversive outcomes. The possible outcomes were big reward, small reward (drop of water), no reward, small punishment, or a big punishment (a puff of air delivered to the animal's face). Each behavioral trial began with a conditioned stimulus (CS; a sound, 1 s), followed by a 0.5-s delay and an unconditioned stimulus (US; outcome). Upon the beginning of training sessions the mice are water deprived. In order to train the mice first to lick and to associate a sound with an outcome, drops of water was dispensed unconditionally and immediately following a sound. This also habituates the mice to an unfamiliar surrounding. Various frequencies were also associated with varying amounts of water dispensed. As the mice learn to lick for water, water

is then dispensed conditionally upon detection of licking. A delay period up to 0.5-ms was slowly introduced so that we could study associative behavior.

When the mice were trained for the reward segment, the punishment trials were introduced consisting of white noise preceding an air puff to the face. In order to better facilitate distinguishing of the appetitive and aversive auditory cues, they were delivered on left and right speakers individually. A lick by the animal closes a circuit and is detected by a lickometer,³⁹ which also detects the rate of licking by the animal during the interval between end of a CS and delivery of a US. Licking rate is one of the parameters by which we gauge how well and how much an animal anticipates an outcome.

Recording Device

As animal subjects, mice provide a very diverse platform for the investigation of behaviors ranging from learning to social performance. Viral expression targeting enables highly precise optogenetic investigation of mouse behavior. The ability to simultaneously record multiple channels of electrical activity during optogenetic manipulation in awake mice has been afforded by the optetrode, which combines electrophysiological recordings of multiple isolated units with optical hardware in awake freely moving mice.⁴⁰ At the heart of the device is an optical fiber, which conducted optical stimulation from a laser. It is surrounded by 16 microwires (channels) wound into four tetrode bundles, which records extracellular signals. These four bundles are separated by a width measuring the diameter of the optical fiber and extend beyond the optical fiber in order to better isolate the signals from individual neurons in the region of interest. These channels allowed us to deconstruct the spikes so that they could be represented in various amplitude spaces. Thus, the fiber also provides structural support for the tetrodes during vertical translation through brain tissue. The fiber-tetrode assembly was combined with a custom mechanical drive that allowed adjustment of depth in the brain region. The tetrode microwires were connected to an adaptor that then amplifies and displays the signal on a recording interface. Following

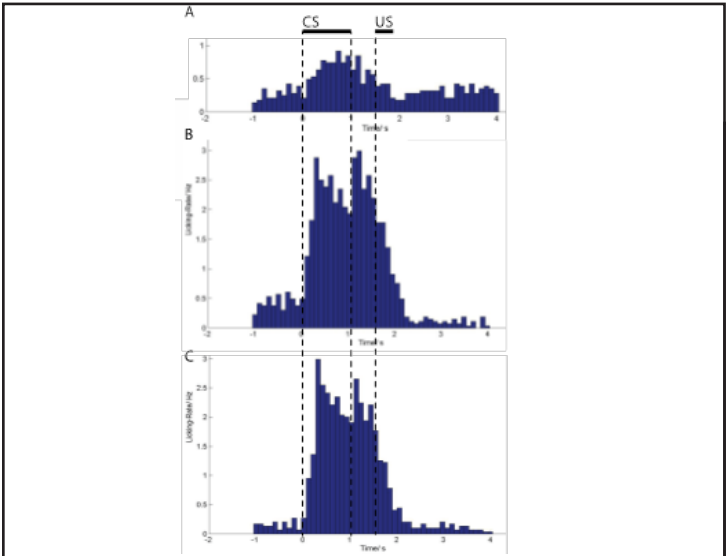


Figure 8: Licking rate during various reward segments. a. No reward. b. Small reward. c. Big reward. Notice licking rate is slightly faster than that in small reward.

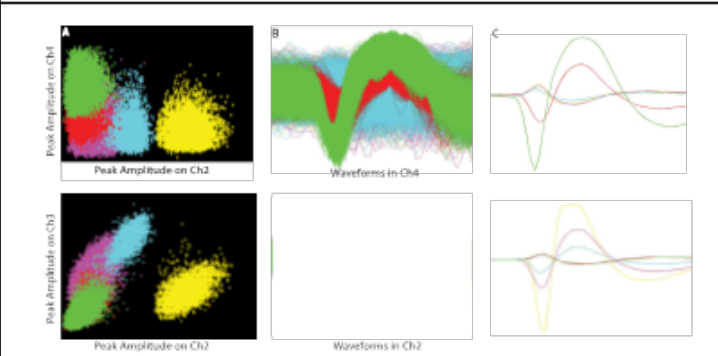


Figure 9: Clustering the spikes. a. Clusters of spikes represented in amplitude space of different channels: good separation between green and red cluster only in one of the channels. b. Actual waveforms of all spikes as seen in various channels. c. Average of the waveforms.

a surgery for implantation, the microdrive was permanently fixed onto the head of the animal. The entire contraption was very light and worn readily on freely moving adult mice, although during the experimental setup the mice were head-fixed, which confers an element of stability to the implanted device.

Spike Sorting

Use of multi-channel electrodes allow for multiple neurons to be recorded simultaneously. Depending on the physical relationship of neurons relative to the multi-channel electrode, the amplitude and extracellular waveform of a neuron on each channel will likely differ from that of a neuron in a different physical location. Clustering is normally accomplished by calculating a set of features of each spike waveform, such as the amplitude on each channel of a tetrode. Spikes presumed to come from the same neuron will form clusters in a high dimensional feature space which can be separated from other clusters representing other simultaneously recorded cells or noise events. Clusters are identified manually or by automatic clustering methods. After clustering spikes into units, it is important to ensure spikes assigned to one cluster are well-separated from other spikes recorded simultaneously, and for that we use two quantitative measures of cluster quality: Lratio and Isolation Distance. The former is an indication of how well separated the clusters are, and the latter is a measure of how well-contained they are. Measurements of isolation of the clusters are especially important in the recording of the globus pallidus because this area exhibits a high baseline activity, and visual clustering of neurons may not be very easy.

The spikes were sorted using a spike-sorting algorithm, MClust.⁴¹ MClust represents each spike as a point in amplitude, energy and wave principle component space as

recorded by each channel. Figure 9 is a rough depiction of clusters taken from tetrode data collected in mice globus pallidus. Some clusters are well-defined and isolated but other clusters required looking in other channels in order to find that they can be better separated. We also found that automatic clustering was oftentimes not very useful as clusters were found to fit the cluster quality parameters instead of the parameters used to judge how well-defined are the clusters, which resulted in very strange (and usually unreliable) clusters. After all spikes have been grouped into clusters from various tetrodes, in different recording regions (brain depths), across a span of several weeks across different training sessions, in different animals, we obtained a good number of functional units. In order to identify the types of neurons present in our region, we look at their activity profile and firing rates when correlated with time points of the CS and US.

RESULTS

Classifying the clusters

We recorded the activity of GPh neurons while mice performed the conditioning task described previously. To characterize the responses of the population, we measured the temporal response profile of each unit (neuron) during the various reward trials by quantifying firing rate. Spike recorded from each channel and tetrode were clustered by their waveform properties, and the spikes in each cluster were then categorized by their preceding CS-reward or punishment- and was subsequently plotted against the time points during the various trials, obtaining first a raster plot of the spike time points (represented by a tick) spanning an entire trial. The tick marks were then binned and plotted as a histogram that represented the mean firing rate across a discrete set of time points during a trial. Henceforth activity of a neuron will be represented as firing rate (number of spikes/sec). Once we

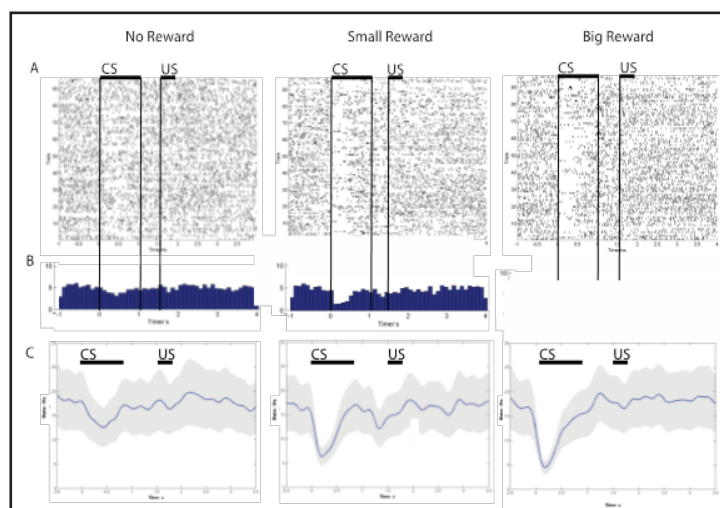


Figure 10: Firing rate during different time points, categorized by no reward, small reward and big reward. The first two panels, A and B, represent a single cluster. a. Raster plot of spikes spanning a trial for multiple trials. b. Histogram of binned spikes. c. Average of firing profile for units in the same tetrode exhibiting similar activity profile. Gray area indicates confidence range.

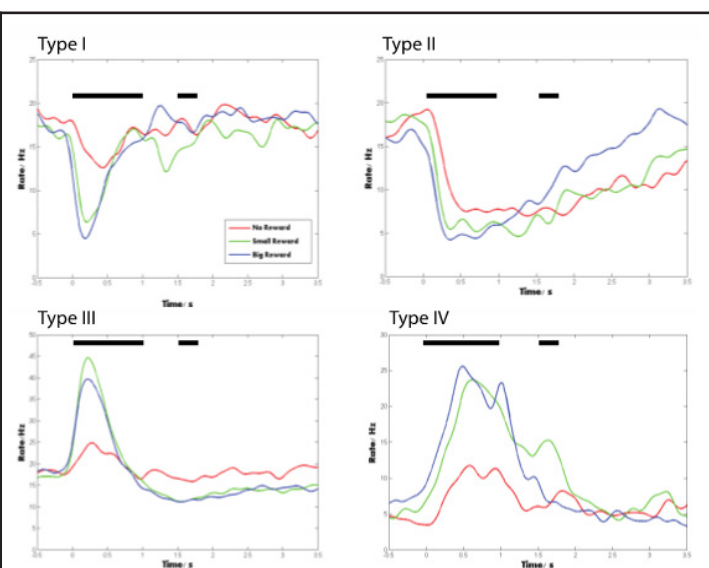


Figure 11: Types of neuron temporal activity profiles present in the recording region. Assigning of type numbers is arbitrary. As before, the long and short black segments indicate CS and US respectively.

obtain the histogram, the activity profile of a unit was average with another unit exhibiting the same activity profile to the same stimulus. One sample trial of a unit is presented. The units that were averaged were taken within the same tetrode, but as more units are collected and analyzed, units from different tetrodes, even different days may be averaged.

After having clustered all the recorded spikes, we can then group the units with the same activity profile. This yields four types of neuronal responses that were distinguished primarily according to the magnitude and length of response to the CS and US. Observing the temporal profiles of responses in trials with rewards, we found neurons that were excited or inhibited phasically by reward or punishment-predicting stimuli. It would be difficult to assign a biochemical character to these neuron types just based on activity profile, especially since the neurons within the GPi that project to the lateral habenula are not very well studied and characterized. With data from the tagging experiments however, we will be able to confirm the biochemical identity of these neurons. Within our recording region, we identified a large number of neurons which showed prolonged inhibition to the CS as well as to the US. Other neurons show a phasic excitation or inhibition to the CS. As the boundaries between the GPi and GPh are not too distinct, it was possible that amongst our recorded neuron population there were neurons from the GPi. The response profile below represents an average of the units which had similar response patterns.

There were other recorded responses which did not facilitate grouping into a distinct category, for which more recordings would be very useful. The majority of the neurons showed a much more pronounced response to the CS (auditory cue) rather than the US (actual reward), although in Type I neurons, there appears to be a slight dip in neuron firing rates preceding the US. Type I neurons are hypothesized to be the ones we are looking for: glutamatergic neurons in GPi that project to the lateral habenula, which exhibit reward negative firing patterns, which is similar to what has been identified.³⁷ These would fit in the picture of the function of these neurons, as depicted in Figure 4: the inactivity of these neurons promotes firing of dopaminergic neurons in the substantia nigra, reinforcing an appropriate motor program. It should also make sense that these neurons also exhibit a certain degree of change in response to the US, as it is the receipt of the US that determines whether a motor program should be reinforced or altered. This consideration may be more important when computing reward prediction errors (RPE).

Reward Prediction Error

An important response property that supports RPE coding in dopaminergic neurons is their decrease in firing rate when an expected reward is omitted.⁴² We omitted reward unexpectedly on about 10% of big reward trials. We also added some trials where a big reward was dispensed when the auditory cue predicted no reward. During the analysis

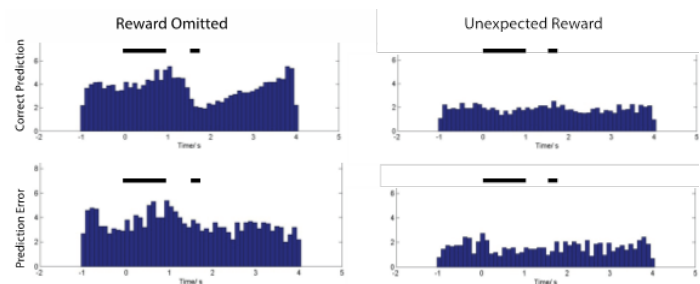


Figure 12: Reward Prediction Error. The panel on the left depicts a reward omission and that on the right depicts an unexpected reward. The reward prediction error response is on the lower panels.

of the clusters, the trials which were marked to be reward prediction error trials were extracted and their response plotted against the time points for CS and US as before and binned and plotted in as a histogram. Very few trials were run and only Type I neurons were analyzed, thus this is just a very preliminary presentation of results of RPE in target GPh neurons.

There is a large difference between the response profiles of correct prediction versus prediction error in the case of reward omission. In this case, the pronounced difference rests in the response to the US, which should be the case as both conditions receive the same auditory cue. It appears that there is an inhibition of the GPh neurons as reward is received, and an absence of inhibition when reward is omitted. When taken together with the dopaminergic neuron data, one can see how this might be a viable way for the globus pallidus to encode RPE with DA neurons in the substantia nigra. As previously established, DA neurons decrease in firing rate when an expected reward is omitted. In Figure 4, inactivity of DA neurons in SNc is correlated with activity in the GPh neurons. As hypothesized, glutamatergic neurons in the GPh activates GABAergic neurons in the lateral habenula, which inhibit the dopaminergic neurons in the SNc. Thus, a decrease in DA neuron activity should correlate with increase activity of the GPh. This is exactly what we see in Figure 12: when reward is omitted, there is a higher firing rate in the Type I GPh neurons as compared with the normal response pattern of a correct prediction of reward. In addition, a correct prediction of a large reward during the US block results in an inhibition of firing of GPh neurons. This phenomenon also agrees with our previous discussion of DA neurons: inactivity of GPh neurons removes the excitatory input to the lateral habenula, which is dis-inhibitory on the DA neurons in the SNc. Thus, DA neurons can release dopamine into the direct and indirect pathway. As previously discussed, this is a reinforcement mechanism which promotes an appropriate motor program resulting in an expected reward, and thus it should not be surprising to see an inhibition of GPh neurons upon a correct prediction.

What about in the case of the unexpected reward? This was only one instance (one cluster) and thus does not afford much in the area of significant differences, but we

can see that the firing rate during the prediction error trials was generally lower and more sparse than that of the correct prediction trials. In this case, the mice hear an auditory cue for nothing, but receive a large reward instead during the instances where we introduced a prediction error. A lower firing rate during the US onset in the prediction error compared with those during a correct prediction would serve the same purpose as that previously discussed: inhibition of GPh is strongly correlated with activation of DA neurons in the SNc, which serves a reinforcing purpose. This is the brain telling the circuits that there might need to be adjustment in the motor program upon hearing the sound for no reward as now there is reward. Thus, the mice might anticipate the no reward signal just slightly more. To summarize, inhibition of GPh neurons indicate a reinforcement of an anticipation (unexpected reward) by allowing the release of DA, whereas activation of GPh indicates a discouragement of an anticipation (reward omission) by inhibiting release of DA, which restates the finding mentioned previously that there is a decrease in DA neurons firing when reward is omitted. This is supported by our preliminary data on reward prediction error. Much more data collection would be crucial to arrive at an observation with greater confidence. Looking at licking rate as a measure of anticipation may also be elucidating.

Reward History

Reward history is essentially a measure of how the value assigned to a particular auditory cue—and thus reward—changed depending on the reward that was received in the preceding trial. It was an attempt to understand (1) if there were any changes (2) and if there were, in which neurons and (3) gain a rough understanding of how reward history was encoded in the neurons in the region of interest. The same experimental procedure as that outlined in classifying the

neuron types was carried out: mice were given auditory cues which always preceded their associated reward and its delivery was conditional upon licking. The only difference was in the data analysis. For each cluster, not only were they separated by the auditory cue for that trial, all the spike time points for each auditory cue was also separated by their preceding outcome: no reward, small reward, big reward. We essentially had nine groups of spikes: 3 different rewards, and each reward following one of the three reward outcomes. Thus it was basically a trick with extracting the relevant information out of the huge amount of data that was collected. We analyzed Type I and Type III neuron firing profiles. Each type is represented by only one cluster as it seemed to be messy to simply average the reward history response profiles for multiple units. Preliminary results are depicted at the bottom of the page

First of all we note that there hardly seems to be any differences in reward history response profiles in Type III neurons. This may come as a slight surprise as a response that reacts so strongly to a CS cue does not have any reward history discrimination, even as it discriminates between the cues for the reward outcomes. The plots for the Type I neuron is slightly more erratic and definitely suffers from lack of a large pool of data to average out the noise. We look first at the no reward cues, and see that there is a significantly larger dip in the CS block for the no reward cues that follow a large reward (blue trace), with a higher baseline activity. Remembering that an inhibition of GPh neurons indicate a greater anticipation, it appears the neurons are telling us that even when the mice know that he is not getting a reward for that trial, because he had a large reward recently, he has a greater anticipation for the current trial. We next look at the large reward cues. Although there is almost no difference between the inhibition of firing during the CS block, we see that there is a lower

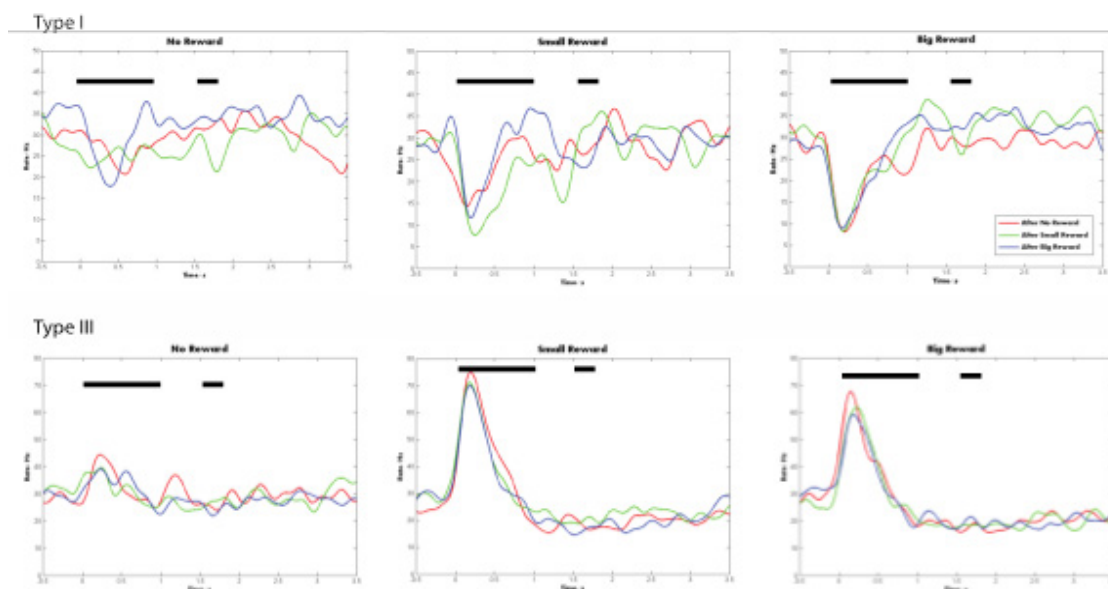


Figure 13: Reward History. As in Figure 11, Type I and Type III neurons response profiles were categorized into reward outcomes. Each reward outcome is further categorized by the outcome they follow.

baseline activity in the big rewards that follow a no reward outcome (red trace). Generally, inhibition of the GPh neurons promotes motor movement, indicating that in this case the mice are slightly more motivated to obtain the reward after not getting anything previously. The higher baseline activity in the neurons in the no rewards cue following a large reward outcome can also be interpreted as the mouse is slightly less motivated after having received a large reward. The small rewards plot does not bear a distinct trend from which we can draw significant conclusions. Reward history does appear to play a role in value-based decision making. It appears that reward history may be encoded in the baseline activity, and to some extent the level of inhibition achieved during a CS block, although that seems more to govern anticipation than motivation, which plays a larger role in reward history. The experimental data matches well with the current model for basal ganglia circuits, but as always, more collection and analysis needs to be done before any conclusions can be drawn.

DISCUSSION

There exists a subpopulation of the GPi neurons that projects to the lateral habenula, termed the GPh neurons. There has been significant evidence pointing to the significance of the lateral habenula in reward learning tasks. The GPi is a major output center in the basal ganglia, commonly associated with learning, reward circuits, motivation, decision-making, but most importantly with motor execution. A decision must be actualized by an action, and thus it behooves us to better understand these GPh neurons as they must serve as a critical link between the motor and limbic circuits. In this series of experiments and data analysis we have attempted first to record from and classify the types of neurons that exist in this region in mice, who have been trained to carry out a set of value-based tasks in which they lick for water. Delivery of water for these water-deprived mice (a reward) is contingent upon the act of licking, and is preceded by an auditory cue (CS) where different frequencies are associated with different outcomes (US: no water, small or large amounts of water). The recording device is similar to the optetrode design, in which an optical fiber is surrounded by 16 microwires wound into 4 tetrode bundles. Waveform properties of each spike event picked up by the tetrodes is represented in a feature space that ultimately allows those with similar properties to cluster together, presumably comprising of all the spikes belonging to a single unit across the entire number of trials. These spikes in each cluster are then separated by the reward outcome in order to observe the responses of different types of neurons in various situations.

We have arrived at least four different types of neurons in our recording region, with phasic and prolonged activation or inhibition to the CS. Type I, with phasic reward negative responses, appears to be the glutamatergic GPh neurons, with an inhibition in response to a cue that predicts reward. This agrees with the circuitry as depicted in Figure 4: inhibition

of the GPi promotes dopamine release and reinforces a lucrative motor program. These neurons also exhibit reward prediction error encoding, although the mechanism is not well-understood. The functional aspect in encoding for RPE appears to be the difference in firing rate during the onset of the US. This would make sense as one would imagine a difference in expectation, or an error in prediction is received when the outcome is different from expected and must be encoded in some form and fed back to the circuit that updates perception and value-based decision making for subsequent trials. We see this phenomenon further in reward history, where the main parameter for motivation after consideration of reward history is in the baseline firing rate of the response profile. Are the elements of the circuit that raises or lowers the baseline and thus motivation following an outcome inherent components of the basal ganglia or are they remote elements in a different, but related circuit?

Other parallel circuits to the motor loop involving the basal ganglia are investigated in the introduction, and increasingly it appears that nature works by analogies and patterns, from homologous limbs to conserved protein folds, to brain circuits. Already we see that analogous elements occupy the same space (Figure 6), and that activation and modulatory mechanisms are similar. What remains is to extrapolate any ideas from a better studied circuit and apply it to parallel circuits, which may facilitate understanding of both circuits. For example, social interactions, which are well studied in the limbic loop, have been known to great counteract the effects of addiction and depression.⁴³ What are the modulatory elements and how does it affect the structures in the basal ganglia? With input and output terminals, effectors that can activate, inhibit or modulate a downstream element, gating mechanisms, and loops that can run in parallel or in series in strategic topography, the neurons of the brain may be able to form a formidable logic circuit that can compute seemingly abstract concepts such as probability, effort, time, payout, etc., which are important considerations for an animal to assign a value to an outcome, and thus make a value-based decision as to whether or not to undertake an action. Thus we can begin to perhaps understand the brain circuitry in the language of the brain: dopamine, instead of being associated with feelings of satisfaction, more directly reinforces an expectation or a motor program, so that we are more likely to repeat the same actions, depression is not sadness, but perhaps more of a lack of motivation due to an overactive LHb or GPh, thus suppressing thalamic motor control, etc. This has strong implications in understanding the process of learning and treating mental disorders. With a basic understanding of electrical circuits an enterprising individual could potentially construct a simplistic functional reward circuit and simulate various reward learning situations, and a blueprint could be especially elucidating of the human brain circuitry. Following this line of thought, regions of the brain must be considered as elements in a complex circuit, rather than individual regions, much less a certain cell type. Thus it would not be very effective

to speak of a certain cell type in the brain, removed from their individual circuits as neurotransmitters are but effectors in a functional loop. One unit does not function alone; it belongs to a network and thus one should be wary of attributing an observed effect completely to a stimulated cell type or brain region, as effects upstream or downstream from that circuit element may not have been duly considered.

There are some experiments yet to be run and analyses yet to be carried out that would make this story more complete. One of the interesting pieces of information could be obtained as early as the first day of training. While for certain pieces of information such as reward prediction error, it is imperative that the animal understands the cues well enough to know that there was supposed to be a reward during the trials when the reward is omitted, during the early stages it would be interesting to see how firing pattern of the neurons in the target region slowly evolves as the mice learn the task. As the reward circuit plays a big role in the process of learning, one can set the detectors to understand how the neurons form connections and ultimately adopt the final response profile. How does it correlate with the learning observed from the anticipatory licking? On the same lines, satiation would also be interesting to look at. We have already seen in the reward history that mice are less motivated after they have had a big reward, but what about when they are satiated up to the point where they would also ignore the big reward? How long before they stop licking does the brain register that it is satiated? Is it simply a change in baseline, thus a lack of motivation, or does the auditory cue not even induce inhibition of firing anymore? To obtain this information regarding satiation and learning, plotting the response across trials, rather than lining them up by trials and plotting over the CS/US time points might be interesting.

Another piece of the puzzle that appears to be blatantly missing is the tagging profile. The experiments were conducted, but the analyses had not been very comprehensive nor conclusive. Using genetic engineering (which is another area where ingenious tools are available to promote better experiment design), we can insert light-gated ion channels such as ChR2 in a specific neuron population (e.g. VGlut2 neurons in the GPi) using a Cre recombinase system. For each neuron, we can measure the response to light pulses and the wave shape of spontaneous and induced spikes. There should be a high correlation between the light pulse and the timing of the spike, and based on the waveform properties we can again represent them in a feature space, and the physical location of the cluster in the same two channel's amplitude space should overlap with a cluster found previously during the electrophysiology recording session. The criterion that the light-evoked waveform must look almost identical to the spontaneous waveform also ensures that the previously identified unit is correctly assigned a biochemical identity. Using retroviral tracing techniques one can also get a better

idea as to the immediate connections to and from a target population of neurons.

Although there were many new techniques that I was exposed to through this project, including murine handling and surgery and building the microdrive, most of my time was spent on data analysis, from clustering to generating raster plots, from data collection to sorting, of a massive amount of information. It would be much more efficient to streamline the entire process of data collection to data analysis so that computations can run in the background while prepping and designing experiments can take place, and freshly collected data is automatically fed through the analysis machinery. Perhaps cleverer experimental design would need to be instituted in order to observe an isolated phenomenon. The brain undergoes and presents a lot of activity and careful extraction and sorting of the data could reveal so much about the brain machinery, and some thought should go into the code that sorts this information. Other analysis methods such as regression coefficients or changes in baseline could more clearly elucidate a trend. The clustering is achieved with MClust which works well most of the time. However, the automatic clustering program KlustaKwik often produces unreliable clusters and most clustering had to be done by hand, which greatly slows down the process. A more reliable automatic clustering software would greatly facilitate the data analysis process.

At last, we note that neural circuits are incredibly complex-and for good reason-but nature uses many analogies and the development of new tools is very promising. Keeping the analogous circuits in mind and using the available tools at hand, we have gotten glimpses of the role that the GPh plays in the basal ganglia circuits: the reward negative neurons which are most likely glutamatergic, encoding for anticipation upon receiving the cue and assigning a value to the associated outcome. Whether it is this value that changes, or the overall baseline motivation that fluctuates when computing reward history remains to be seen as more units are recorded and analyzed, which will also elucidate the changes in temporal firing rate when the neurons are encoding reward prediction error. For now the reward evaluation pathway consisting of the glutamatergic GPh neurons, GABAergic LHb neurons and DAergic SNc neurons stands up to experimental data, however how they interact to galvanize a decision and what roles they play in other circuits still remains to be studied.

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it's taken me a long time to comprehend what he meant but slowly I am beginning to arrive at my own understanding. My own personal interest in making tools is I find more and more supplemented by my understanding of the population that needs it and the purpose it will serve.

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The Human Microbiome: Slowly Getting There

By Alexander Reynaldi Posted On November 9, 2015

At this point in time, the study of the human microbiome is not a novelty. Quite a lot of time and money has gone into pursuing the promising field, hoping that collecting data from the trillions of microorganisms in and on our bodies will offer insights into how they affect health and diseases. While the microbiome has been shown to heavily affect us—the food we eat, our immune system and infections, organ developments, even behavioral traits—our knowledge regarding the microbiome is still extremely limited. The goal of predicting an individual's propensity for certain diseases (and ultimately preventing them) using the human microbiome seems more distant than not.

Part of the reason of why this research seems to be progressing slowly is the vast amount of data that needs to be processed and the time required to amass it. Specifically, months are required for bacteria collection (mainly from feces—relatively unappealing to the masses and probably another reason the field is not popular) and for gene sequencing. Biotech companies such as Biomiic have started working on how to process and present collected data at a much faster rate.

Once data can be processed more powerfully, perhaps the field will advance rapidly. After all, even the world's largest collaborative biological project—The Human Genome Project—was only possible because of remarkable progress in sequencing and computing technology.

In any case, the study of the human microbiome is extremely valuable as our microbiome is an integral part of our lives. Perhaps once it gains more popularity and funding, more will be discovered regarding these organisms that call us their

Scientists Selling Genetically-Engineered Micro-Pigs

By Kara Turner Posted On October 19, 2015



Who doesn't love things that are fun-sized? While most pet owners would gladly keep their furry friends baby sized forever, a group of scientists in China has taken things a step further. Geneticists from leading genomics research institute BGI in Shenzhen, China have begun selling genetically engineered micro-pigs as pets starting at US\$1600.

By deactivating a growth hormone receptor or GHR gene, scientists have effectively stunted the growth of Bama pigs. Normally mature pigs weigh up to 100 pounds, but mature micro-pigs grow to only about 30 pounds, or the size of an average dog. By introducing an enzyme called transcription activator-like effector nucleases, or TALENs, to the cloning process, scientists were able to disable one of two growth hormone genes that cause Bama pigs to mature to their full size.

Of course, cloning Bama fetuses comes with adverse health effects and shortened lifespan, as evidenced by other cloned mammals, such as Dolly the sheep. However, by breeding the genetically engineered male micro-pigs with normal female pigs, half of the offspring are born as micro-pigs without the adverse health effects of being born as clones.

Having more similar genetic and physiological make-up to humans than the typical lab rat, but often rejected for lab work for their large size, micro-pigs were originally intended to serve as subjects for human disease in genetic research. However, a fringe pet market for unusually small animals has given their products new purpose. As of now, BGI states that profit is currently their main objective with their new micro-pigs.

IMAGE SOURCES

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<http://6.darkroom stylist.co.uk/980/3ae5cae2e-2b91a857e4e5900fb0083b5:2d4802eda6893fb3a5563a36d-b66e9b5/micro-pigs-3-rex-18apr12.jpg>

